

REASONING & COMPUTER APTITUDE

1. In the question given below, three conclusions are given followed by five statements. You have to choose the correct set of statements that logically satisfies the given conclusions disregarding the commonly known facts.

Conclusions:

- I. Some atoms being matter is a possibility.
- II. Few elements are not particles.
- III. No matter is molecules.

Statements:

- I. Every molecules are atoms. No matter is particle. Only a few elements are matter. Some particles are not molecules.
 - II. None of the particles are molecules. Half elements are molecules. All matter are atoms. Some atoms are particles.
 - III. Not a single matter is particle. Only a few elements are matter. All atoms are elements. Very few molecules are particles.
 - IV. All molecules are Atoms. All matter are atoms. Some particles are elements. No element is molecule.
 - V. Some particles are molecules. Only a few matter are elements. No particle is atom. Each atom is element.
- (a) All statements satisfy
 - (b) Both statements II and V
 - (c) Only statement III
 - (d) Both statements I and IV
 - (e) None of the statements satisfy

Directions (2-6): Read the following passage and answer the given questions.

A word and number sorting machine, when given an input of words and numbers, rearranges them in a particular manner step by step, as indicated below, until all the words and numbers are arranged. Given below is an illustration of this arrangement.

Input: 37 50 FGN KZU 74 YNS 51 100 ORS 17 HPBM

Step 1: 100 37 50 FGN KZU 74 YNS ORS 17 HPBM 51

Step 2: 74 100 50 FGN KZU YNS ORS 17 HPBM 51 37

Step 3: 50 74 100 FGN KZU YNS ORS HPBM 51 37 17

Step 4: 50 74 100 EFGNM KZU YNS ORS HPBM 51 37 17

Step 5: 50 74 100 EFGNM GHPBML KZU YNS ORS 51 37 17

Step 6: 50 74 100 EFGNM GHPBML JKZUT YNS ORS 51 37 17

Step 7: 50 74 100 EFGNM GHPBML JKZUT NORSR YNS 51 37 17

Step 8: 50 74 100 EFGNM GHPBML JKZUT NORSR XYNSR 51 37 17

And, step 8 is the last step of the above input, as per the rules followed in the steps given above; find out in each of the following questions the appropriate step for the given input.

Input: ETU 75 GTQ 14 RTV 22 NTS 32 MTW 67 FSN 19 CER

2. How many elements are there between 'RTV' and 'NTS' in the step prior to the penultimate step?
 - (a) None (b) One (c) Two
 - (d) Three (e) Four
3. How many elements are there in between 'BCERQ' and 75 in the step 9?
 - (a) 5 (b) 4 (c) 8
 - (d) 6 (e) 3

4. What is the position of 'EFSNM' in the step 7?
 - (a) Seventh from the left (b) Seventh from the right
 - (c) Fifth from the right (d) Sixth from the left
 - (e) None of these
5. Which element is second to the left of 'GTQ' in the step 4?
 - (a) BCERQ (b) RTV (c) MTW
 - (d) 75 (e) None of these
6. Which of the following is the last step?
 - (a) 7 (b) 9 (c) 11
 - (d) 8 (e) 10

Directions (7-10): Study the following information carefully to answer the given questions.

An uncertain number of persons are sitting in a straight line facing the north direction. The persons sitting at the odd-numbered positions are Managers and the persons sitting at the even-numbered positions are Associates in different companies.

Both X and Z are seated to the left of U and P, but to the right of Q. There are five persons between J and G. The person sitting fifth to the right of Q is an Associate. Two persons are seated between Q and R. The number of persons sitting between Q and the Manager at the extreme left end is one more than the number of persons sitting between J and P. The number of persons sitting between R and G is the same as the number of persons sitting between X and Z. Four persons are seated between Q and T, who is not seated to the right of R. The Manager at the extreme left end of the row is S. Three Managers are seated between T and X. J sits 3rd to the right of R. The number of persons sitting between Q and R is one more than the number of persons sitting to the right of G. P is the third Associate to the right of J.

7. How many persons are seated between the third Associate to the right of J and the fourth Manager from the extreme right end of the row?
 - (a) 4 (b) 5 (c) 9
 - (d) 11 (e) More than 11
8. Who is seated exactly in the middle of R and the person who sits immediately to the left of Z, if the total number of persons is known and the seating alternates between Managers and Associates?
 - (a) Q (b) X (c) J
 - (d) Unknown person (e) G
9. Which of the following statements is definitely true?
 - (a) The number of persons sitting to the right of Z is equal to the number of persons sitting between Q and R.
 - (b) T is seated to the right of X.
 - (c) S is seated to the right of J.
 - (d) The number of Managers sitting between X and P is more than two.
 - (e) The person is seated 5th to the right of Q is P.
10. What is the position of the one who sits third to the left of the 5th Manager from left end of the row with respect to Q?
 - (a) Third to the right (b) Immediate left
 - (c) Second to the left (d) Fifth to the right
 - (e) Fourth to the left
11. In the question some symbols —, #, /, ¥, β and * are used with the following meaning as illustrated below:
 - 'P:Q' means 'Q is father of P'
 - 'P#Q' means 'P is mother of Q'

'P/Q' means 'P is sister of Q'
 'P¥Q' means 'Q is brother of P'
 'PβQ' means 'P is wife of Q'
 'P*Q' means 'Q is wife of P'

Which of the following expressions mean L is sister-in-law of X?

- (a) X¥H:C¥L (b) L/XβT¥E
 (c) L/X¥F ¥G (d) L β G ¥X*C
 (e) None of these

Directions (12-15): Each of the questions below, consist of a question and three statements numbered I, II and III. You have to decide whether the data provided in the statements are sufficient to answer the question. Read the three statements and give answer:

- (a) Data in statements I, II and III together are sufficient to answer the question.
 (b) Data in statements I, II and III together are not sufficient to answer the question.
 (c) Data in statements I and II together are sufficient to answer the question.
 (d) Data in statements I and III together are sufficient to answer the question.
 (e) Data in statements II and III together are sufficient to answer the question.
12. Six persons sit around a triangular table. Three persons sit at each corner of table and rest sit at middle of each side of table. All of them face inside and like different colours. Who among the following likes Green colour?
 I. Y likes Purple and sits second to the left of X. W is not an immediate neighbor of X but sits third to the right of the one who likes Pink colour.
 II. The one who likes Pink sits second to the right of Z, who does not sit at any side of the table. U neither likes Black nor Green. The one who likes Green sits second to the right of the one who likes Black colour.
 III. V likes White colour and sits second to the left of the one who likes Yellow colour.
13. A certain number of persons sit in a line facing north. Then, how many persons sit between P and V?
 I. P sits 3rd to the right of Q, who sits 4th from extreme end. Q is an immediate neighbour of both R and S.
 II. T sits 2nd to the left of U, who sits 3rd from the extreme end. There are as many persons sit between V and U as between U and W.
 III. Three persons sit between R and W. X sits 2nd from extreme left end. T sits between P and W.
14. Eight persons i.e. A, B, C, D, E, F, G and H sit around a circular table facing inside, but not necessarily in the same order. Then, who sits 2nd to the left of B?
 I. Three persons are sitting between A and B, who is an immediate neighbour of C. D faces E.
 II. F is an immediate neighbour of E and A, who sits 3rd to the left of H.
 III. G faces H, who is an immediate neighbour of B. C is not an immediate neighbour of D.
15. What are the words for code "dt ef" as per the given code language?
 I. "ninja turtle game famous" is coded as "cd ef dk mn"
 II. "game in world the" is coded as "dk pq lm dt",
 "ninja in the world" is coded as "pq lm dt cd"
 III. "turtle now the world" is coded as "lm ex dt mn"
16. In the word 'COLLABORATION', all consonants are replaced with their preceding letters, and all vowels are replaced with their succeeding letters and in the word 'DETERMINATION', letters appearing before N in the English alphabetical series are replaced with their succeeding letters, while letters appearing after N are replaced with their preceding letters and N remains as it is.

After that, combine the transformed letters from both words into a single sequence, arrange them in alphabetical order. Now, determine which of the following letter is 3rd to the right of the 2nd letter from left end?

- (a) J (b) F (c) B
 (d) N (e) A

Directions (17-18): Read the following information carefully and answer the given questions.

'G % H' means 'G is the husband of H'.
 'G @ H' means 'G is the wife of H'.
 'G # H' means 'G is the sister of H'.
 'G * H' means 'G is the brother of H'.
 'G = H' means 'G is the mother of H'.
 'G + H' means 'G is the father of H'.
 'G ^ H' means 'G is the son of H'.
 'G ≠ H' means 'G is the daughter of H'.

Example: G 2% H means G is the husband of H and G has two siblings.

The following is the family of Y, a married person:

N 2+ P, O 1% D, A 0@ N, D 0= T, C 2# T, M 1^ A, Y 1* O, S 0≠ B, E 1* R, F 0^ R, R 1% C

17. How S is related to O?
 (a) Daughter (b) Sister-in-law
 (c) Niece (d) Daughter-in-law
 (e) None of the above
18. How is N related to R?
 (a) Brother-in-law (b) Father-in-law
 (c) Father (d) Cousin
 (e) None of the above

Directions (19-22): Read the following information carefully and answer the given questions.

Eight persons — A, B, C, D, E, F, G, and H are sitting around a circular table facing toward the center, but not necessarily in the same order. Also, each of them is holding number cards from 2 to 9 (one card per person). The numbers on their cards are distinct. Consecutive numbered card holders are not sitting adjacent to each other.

B is sitting second to the left of the person holding the card numbered 7. The person holding the card numbered 3 is sitting second to the right of E, who holds an even-numbered card greater than 4. C holds a card whose number is a prime number, but C is not sitting adjacent to either B or the person holding the card numbered 9. The person holding the card numbered 9 is sitting immediately to the right of A, who holds a card number divisible by both 2 and 4. The difference between the numbers on the cards hold by D and the person sitting opposite D is 3. The person holding the card numbered 2 sits third to the left of the person holding the card numbered 6. The sum of the numbers on the cards hold by G and the person sitting to G's immediate right is 11. The person holding the card numbered 4 is sitting opposite the person holding the card numbered 8.

19. Which of the following is true about the seating arrangement?
 (a) The person holding card numbered 6 sits second to the right of A.
 (b) C is seated opposite to the person holding card numbered 9.
 (c) The person holding card numbered 4 sits third to the left of E.
 (d) G and the person holding card numbered 3 are seated adjacent to each other.
 (e) None of the above.
20. Who is seated immediately to the left of the person holding the card numbered 7?
 (a) A (b) F (c) B
 (d) G (e) Can't be determined
21. What is the card number held by the person seated opposite to B?
 (a) 8 (b) 6 (c) 4
 (d) 9 (e) 2

22. If the sum of the card numbers held by A and H is 7, then what will be the difference between the card numbers held by C and the one who is sitting second to the left of E?
- (a) 0 (b) 2 (c) 4
(d) 3 (e) 1

Directions (23-24): Read the following arrangement of symbols, letters, and numbers carefully and answer the questions based on it.

& F A # L K 4 \$ 9 G P @ 8 B S M @ € 5 D % L β K 4 A 9 @ S W ¥ 6 B & 8 # J

Step I: All the symbols, which are immediately followed by a letter, will be placed from extreme left end in the same order.

Step II: If a letter is immediately followed by a symbol, then such letters are placed from extreme right end in the English Alphabetical order.

Step III: All the numbers between the 7 element from the left end and 7 element from the right end are placed between the 11 and the 12 element from the left end in descending order.

Step IV: The first ten elements of the arrangement from the left end are placed from extreme right end in the same order.

Note: Step II is performed after the completion of Step I, similarly, Step III is performed after the completion of Step II, and so on.

23. How many elements are between two 'A' after applying step IV in the given arrangement?
- (a) 13 (b) 17 (c) 14
(d) 15 (e) 16
24. Which element is 7 to the left of 21 element from the left end after, applying step IV in the given arrangement?
- (a) S (b) L (c) D
(d) € (e) @

Directions (25-26): Read the following passage and answer the given questions.

Microsoft and OpenAI have forged a strategic alliance to advance the frontier of artificial intelligence (AI) research and development. Leveraging Microsoft's vast resources and technological prowess alongside OpenAI's cutting-edge AI research, the partnership aims to accelerate innovation in AI technologies. Together, they collaborate on projects aimed at tackling complex challenges across various domains, from healthcare and climate change to education and beyond. This collaboration underscores their shared commitment to harnessing the power of AI for the betterment of society.

25. Which of the following statement can be concluded from the passage?
- Microsoft and OpenAI have joined forces to pioneer advancements in artificial intelligence (AI) research and development.
 - The partnership between Microsoft and OpenAI combines Microsoft's extensive resources and technological prowess with OpenAI's cutting-edge AI research.
 - The partnership between Microsoft and OpenAI focuses solely on developing consumer-oriented AI products and services.
- (a) Only I and II (b) Only II and III
(c) Only I and III (d) Only II
(e) Only III
26. Which of the following statement should be the course of action?
- Expand the scope of collaborative projects between Microsoft and OpenAI to include research initiatives aimed at addressing global challenges such as climate change, healthcare disparities, and education access.
 - Transition Microsoft and OpenAI's partnership to focus exclusively on commercializing AI technologies for profit-driven ventures.
 - Discontinue collaborative efforts between Microsoft and OpenAI due to conflicting research priorities and organizational objectives.

- (a) Only I (b) Only II (c) Only I and II
(d) Only I and III (e) Only II and III

27. We have five words given below, such that one letter of each word is bold. Find how many meaningful words can be formed by using the letters (only once) which are bold in each given word?

A. Wonderful B. Journey
C. Technology D. Famous
E. Inspiration

- (a) One (b) None (c) Two
(d) Three (e) More than three

Directions (28-32): Read the following information carefully and answer the given questions.

Six persons: A, B, C, D, E, and F were born on different dates in March 2000, falling between the 20th and 28th, but not necessarily in the same order. There are three vacant dates, and these are not consecutive.

A was born on an odd-numbered date. B was born on the 25th, but not before C. Three persons were born between A and F. F was born just next date of E. There are two vacant dates between C and F. D was born on an even-numbered date, and was born neither just before E nor just after A.

28. Which of the following pairs of dates could be vacant, based on the given information?
- (a) 21st and 24th (b) 22nd and 27th
(c) 23rd and 26th (d) 20th and 28th
(e) 24th and 26th
29. On which of the following dates was D born?
- (a) 21 (b) 20 (c) 26
(d) 24 (e) 25
30. How many people were born between B and C?
- (a) One (b) Two (c) Three
(d) None (e) Four
31. Which of the following statements is true?
- (a) E and B were born on consecutive dates.
(b) A was born three days after F.
(c) D was born before C.
(d) D was born on the 25th.
(e) All are true
32. Who among them was born on 22nd March?
- (a) A (b) D (c) C
(d) No one (e) E
33. In the question below, a few statements are given depicting an issue or a problem. You have to assume everything in the statements to be true even if they are at variance from commonly known facts, and on the basis of the information given in the statements, decide which of the suggested courses of action logically follow(s) for pursuing.
- Statement:** Semiconductor facilities play a critical role in the global electronics industry's supply chain.
- Courses of Action:**
- Governments should impose heavy tariffs on semiconductor imports, hindering the global electronics industry's access to critical components.
 - Manufacturers should neglect investments in research and development, relying solely on outdated semiconductor technologies for production.
- (a) Only I follows (b) Only II follows
(c) Either I or II follows (d) Neither I nor II follows
(e) Both I and II follow

34. The question given below consists of a statement, followed by two arguments numbered I and II. You have to decide which of the arguments is a 'strong' argument and which is a 'weak' argument.

Statement: Should the RBI lift the ban on the BOB World app?

Arguments:

- I. Yes, lifting the ban promotes financial inclusion, but it may also risk exposing users to potential security threats.
 II. No, maintaining the ban ensures financial security and protects users from potential fraudulent activities and data breaches.
- (a) If only I is strong (b) If only II is strong
 (c) If either I or II is strong (d) If neither I nor II is strong
 (e) If both I and II are strong

35. In the following question, there is a statement followed by two assumptions labeled I and II. Examine the statement and determine which of the assumptions is implicit.

Statement: Immigration policies should balance national security concerns with humanitarian values, promoting fair and orderly migration processes, fostering social integration, and addressing economic needs.

Assumptions:

- I. Stricter immigration controls are necessary to ensure national security and manage the influx of migrants, while upholding legal immigration pathways.
 II. Welcoming and inclusive immigration policies benefit societies by enriching cultural diversity, filling labour gaps, and contributing to economic growth and innovation.
- (a) Only assumption I is implicit
 (b) Only assumption II is implicit
 (c) Either I or II is implicit
 (d) Neither I nor II is implicit
 (e) Both I and II are implicit

Directions (36-39): Study the following information carefully and answer the given questions.

Five ships (A, B, C, D, and E) are moored at Dock M, and their captains (P, Q, R, S, and T) are steering their respective ships. Each ship starts from the dock, moves in a specific direction, and covers a certain distance before reaching its final position which is named as ship's captain's name.

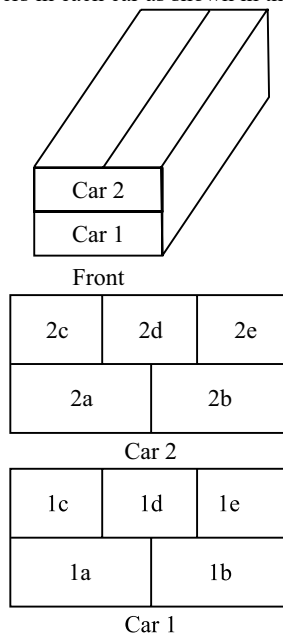
Movements of the ships:

- Ship A, steered by Captain P, starts moving north and covers 5m, then turns left and moves 8m, finally turning right and moving 10m to reach its final position.
 - Ship B, steered by Captain Q, moves east for 12m, then turns right and moves 7m, finally turning right again and moving 3m to its final position.
 - Ship C, steered by Captain R, starts by moving south for 6m, then turns right and moves 5m, then turns left and moves 9m to its final position.
 - Ship D, steered by Captain S, moves west for 10m, then turns left and moves 4m, finally turning right and moving 6m to its final position.
 - Ship E, steered by Captain T, starts moving north for 8m, then turns right and moves 10m, finally turning right again and moving 5m to its final position.
36. In which direction is S with respect to R in the final position?
 (a) North (b) North-west
 (c) North-east (d) South-west
 (e) South
37. What is the total distance covered by Q and R in their journey?
 (a) 32m (b) 22m (c) 62m
 (d) 42m (e) Can't be determined
38. Which of the following statements is true?
 (a) P is in north-east of Q.
 (b) Shortest distance between P and R is 19m.
 (c) T is in north-east of Q.
 (d) Ship E covered 33m total distance.
 (e) All are true
39. What is the shortest distance between the initial and final position of ship B?

- (a) 130 m (b) $\sqrt{100}$ m (c) 120 m
 (d) $\sqrt{130}$ m (e) None of these

Directions (40-41): Study the following information carefully and answer the given questions.

There are two cars, Car 1 and Car 2, with 9 persons (A to I) traveling. One car has 5 persons, and other car has 4 persons. All of them are facing south as seated in each row of each car. Each car has three seats at the back and two seats at the front, such that one of the back seat in one car is vacant. Also, consecutively named persons are not sitting adjacent to each other (neither beside/front/back nor above/below). Car 2 is placed above the Car 1. The seats are arranged in two rows in each car, with the following seat numbers in each car as shown in the diagram:



A is seated in Car 1, either in seat 1a or 1b. No one is seated between B and E. No one is seated behind E. F is seated in Car 2 and in the same row with B. I is just diagonally above C. No one is to the left of G. D is not seated adjacent to I. The seat just behind C is not vacant. D is not seated in the middle of the row of seats.

40. Who among the following is seated in 1d?

- (a) H (b) D (c) G
 (d) No one (e) F

41. Four of the following five are alike in a certain way and thus form a group, then who among the following doesn't belong to that group?

- (a) F (b) B (c) C
 (d) E (e) G

Directions (42-45): Read the following information carefully and answer the given questions accordingly.

Eighteen students are divided into six different groups viz., Red, Violet, Brown, Orange, Purple and Olive. They are sitting in a linear row and facing north direction. Each group consists of three students. A, B, C, D, E and F are captains of their groups. The distance between any two adjacent groups is equal. Other group members are G, H, I, J, K, L, M, N, O, P, Q and R.

G, C and P belong to Violet group. Two groups are sitting between Brown and Purple groups. Neither Brown nor Purple group sits at either end. Orange group sits between Brown and Violet groups. Red group is third to the left of Olive group. A and D belong to Olive and Orange group, respectively. D sits to the left of C. Nine members are sitting between B and E, who belongs to Purple group. F is captain of O and M. K and I belong to the same group, but neither Orange nor Purple group. H and L belong to the group that is to the left of D. R is not a member of

Purple group. J and E do not belong to the same group. B doesn't belong to Brown group.

42. How many students are seated to the right of group member O?
(a) Three (b) Six (c) Nine
(d) None (e) More than nine
43. B and H belong to the _____
(a) Brown group
(b) Red group
(c) Purple and Red groups respectively
(d) Olive and Brown groups respectively
(e) None of these
44. Which among the following options is the odd from the rest?
(a) K and I are in Olive group
(b) O and M are in Brown group
(c) J and N are in Orange group
(d) E and Q are in Purple group
(e) B and L are in Red group
45. Which of the following group/groups is/are seated between Red group and Violet group?
(a) Both Brown and Purple groups
(b) Olive group
(c) Orange group
(d) None
(e) All three (a), (b), and (c)

ENGLISH LANGUAGE

Directions (46-54): Read the following passage and answer the given questions.

Japan is at the forefront of a demographic shift that is reshaping societies worldwide. With one of the most rapidly ageing populations globally, it faces unique challenges and opportunities in maintaining economic vitality and social stability. The Japanese government has developed a multifaceted approach to address these issues, encompassing healthcare, labor market reforms, and community-based initiatives.

As of recent estimates, over 29% of Japan's population is aged 65 or older, making it the world's oldest society. The country's low fertility rate and high life expectancy contribute to this demographic phenomenon. While increased longevity is a testament to advancements in healthcare and living standards, the shrinking working-age population poses economic challenges. Dependency ratios are rising, with fewer workers supporting a growing number of retirees. Healthcare is a cornerstone of Japan's ageing population strategy. The country operates a universal healthcare system that provides affordable and high-quality medical services. However, the growing number of elderly individuals has strained resources. To address this, the government has introduced policies to shift focus from hospital-based care to community-based care. This model emphasizes home healthcare services, enabling older individuals to receive medical attention and support in familiar surroundings. Technological innovation plays a significant role in improving elderly care. Japan has been a pioneer in developing and deploying robotic assistants for caregiving. Robots equipped with artificial intelligence (AI) help with tasks such as lifting patients, providing companionship, and monitoring health conditions. These advancements not only alleviate caregiver burdens but also ensure the elderly can maintain a degree of independence.

The ageing population has also prompted Japan to rethink its labor policies. To mitigate the economic impact of a shrinking workforce, the government is encouraging older individuals to remain employed longer. In 2021, Japan raised the mandatory retirement age for public servants to 65, with further plans to allow workers to continue until 70. Many private companies have followed suit, creating flexible work arrangements to accommodate senior employees. Additionally, the government is actively promoting women's participation in the workforce, addressing

a long-standing gender gap. By enhancing access to childcare and promoting equal pay initiatives, Japan aims to unlock the potential of its underutilized human resources. Efforts are also underway to attract foreign workers, albeit at a cautious pace, given Japan's historically conservative approach to immigration. Policies such as the Specified Skilled Worker visa program aim to fill labor shortages in key sectors like healthcare and construction. Japan's pension system faces sustainability challenges due to the increasing ratio of retirees to active contributors. To ensure its viability, the government has implemented several reforms, including raising the pension eligibility age and encouraging private retirement savings. Additionally, public awareness campaigns emphasize the importance of financial planning for old age.

Recognizing the importance of social inclusion, Japan has also invested in community-building initiatives to combat the isolation often experienced by the elderly. Programs such as "Silver Human Resource Centers" offer older individuals opportunities to engage in part-time work or volunteer activities, fostering a sense of purpose and community. Intergenerational programs that bring together youth and seniors are also gaining popularity, promoting mutual understanding and support. Japan has embraced smart city concepts to cater to an ageing society. Urban planning now incorporates features like barrier-free public spaces, accessible transportation systems, and community hubs for elderly residents. Technology-driven solutions such as health monitoring apps and remote consultations have become widespread, ensuring older individuals can easily access healthcare and support services. Despite these efforts, challenges remain. The cost of maintaining an ageing society is immense, and Japan's public debt is among the highest globally. Balancing the needs of the elderly with those of younger generations requires careful policy-making and equitable resource allocation. Furthermore, cultural attitudes towards immigration and gender roles must evolve to fully address labor shortages. Japan's proactive approach to its ageing population provides valuable lessons for other nations facing similar demographic trends. By investing in healthcare, technology, and inclusive policies, Japan demonstrates how an ageing society can adapt and thrive. While challenges persist, the resilience and innovation embedded in Japan's administrative strategies offer hope for a balanced and sustainable future.

This comprehensive approach positions Japan not only to manage its demographic challenges but also to serve as a global leader in addressing the complexities of an ageing world.

46. What can be inferred as the primary reason for Japan's ageing population challenges?
(a) High rates of immigration
(b) Decreasing fertility rates combined with increasing life expectancy
(c) Lack of technological innovation in healthcare
(d) Overdependence on foreign workers
(e) Declining investments in urban planning
47. Why is Japan encouraging community-based healthcare instead of hospital-based care for the elderly?
(a) To reduce healthcare costs and resource strain
(b) To increase hospital revenue
(c) To eliminate dependency on robotic caregivers
(d) To privatize healthcare services
(e) To introduce stricter healthcare policies
48. What is a plausible reason for Japan promoting the participation of older individuals in the workforce?
(a) To maintain traditional labor practices
(b) To offset the economic impact of a shrinking workforce
(c) To discourage younger workers from emigrating
(d) To attract more foreign investments in technology
(e) To align with global retirement age trends
49. Which policy reflects Japan's cautious approach to immigration?
(a) Raising the retirement age to 70
(b) Promoting gender equality in the workforce
(c) The introduction of the Specified Skilled Worker visa program

- (d) Launching universal healthcare
- (e) Creating intergenerational programs

50. Why is technology significant in Japan's elderly care reforms?
- (a) It helps reduce hospital construction costs
 - (b) It replaces the need for community-based programs
 - (c) It aids caregivers and supports elderly independence
 - (d) It focuses solely on financial planning for retirees
 - (e) It eliminates the need for intergenerational initiatives
51. What inference can be made about Japan's pension reforms?
- (a) They aim to completely eliminate public pensions.
 - (b) They are intended to reduce taxes for retirees.
 - (c) They encourage individuals to rely more on personal savings.
 - (d) They aim to increase benefits for the elderly.
 - (e) They replace the need for financial planning.
52. Why has Japan invested in intergenerational programs?
- (a) To combat isolation among the elderly and promote mutual understanding
 - (b) To encourage older people to mentor younger workers
 - (c) To reduce the dependency on immigration policies
 - (d) To ensure equal employment opportunities for all age groups
 - (e) To train the elderly for technology-based jobs
53. What does the term "Silver Human Resource Centers" imply about Japan's approach to its ageing population?
- (a) Japan focuses only on full-time jobs for the elderly.
 - (b) The elderly are encouraged to engage in community work or part-time employment.
 - (c) It signifies Japan's plan to reduce public spending on senior citizens.
 - (d) The government prioritizes pension reform over employment.
 - (e) These centers are intended to replace retirement homes.
54. What challenge does Japan still face despite its proactive measures for the ageing population?
- (a) Lack of technological advancements
 - (b) Insufficient public debt management
 - (c) Inadequate healthcare for the elderly
 - (d) Over-reliance on immigration policies
 - (e) Limited focus on gender equality

Directions (55-60): Read the following passage and answer the given questions.

The proliferation of online advertising for betting apps has significantly influenced young audiences, often with troubling implications. These advertisements, leveraging sophisticated digital marketing strategies, have become pervasive across social media, streaming platforms, and gaming apps. While the allure of these ads lies in their flashy visuals, promises of quick wealth, and celebrity endorsements, the effects on impressionable youth are multifaceted, raising concerns about their mental, social, and financial well-being.

One of the most pronounced effects is the normalization of gambling behavior. Betting apps often frame gambling as a harmless or even glamorous pastime, obscuring its risks. Advertisements rarely emphasize the addictive nature of gambling or the potential for financial loss, leading young individuals to perceive it as a risk-free opportunity. This normalization can desensitize them to the broader societal and personal harms of gambling, fostering habits that may persist into adulthood.

Psychologically, these advertisements exploit cognitive biases like the "illusion of control" and the "gamblers' fallacy," wherein users believe their actions or luck can influence outcomes. For youth, who are still developing critical reasoning skills, such messages can be particularly compelling, increasing the likelihood of impulsive decision-making. Furthermore, the integration of gamified elements, such as streaks or rewards, mimics the mechanics of video games, making betting apps more appealing and addictive to younger audiences.

Socially, the impact of betting app advertisements manifests in strained relationships and peer pressure. Youth may face pressure to

participate in betting as it becomes normalized among their social groups. This can also lead to conflicts within families, especially when financial losses accumulate or academic performance declines due to preoccupation with gambling.

Financially, the consequences are severe. Youth often lack stable income sources and financial literacy, making them vulnerable to significant losses. Many are lured by promotions like free bets, only to find themselves caught in a cycle of chasing losses.

In the absence of stricter regulations on online advertising, these effects are likely to intensify. Policymakers and educators must address this issue by raising awareness, enforcing stricter advertising standards, and promoting responsible behavior to shield young minds from the predatory allure of betting app promotions.

55. Why are betting apps particularly appealing to youth, according to the passage?
- (a) They offer guaranteed monetary returns.
 - (b) They replicate the mechanics of video games to increase engagement.
 - (c) They emphasize the importance of financial literacy.
 - (d) They are endorsed by family and educational institutions.
 - (e) They focus on long-term investment strategies.
56. What inference can be drawn about the portrayal of gambling in online betting advertisements?
- (a) It is presented as a serious financial activity requiring expertise.
 - (b) It emphasizes the addictive nature of gambling.
 - (c) It downplays risks and highlights gambling as a glamorous activity.
 - (d) It focuses on the social harms associated with gambling.
 - (e) It aims to discourage impulsive decision-making among users.
57. What is the primary psychological tactic used by betting advertisements to attract youth?
- (a) Promoting financial literacy and responsibility.
 - (b) Exploiting cognitive biases such as the "illusion of control."
 - (c) Highlighting long-term economic benefits.
 - (d) Encouraging teamwork and collaboration.
 - (e) Demonstrating the negative consequences of gambling.
58. What social consequence is implied for youth who are influenced by betting app advertisements?
- (a) Improved relationships with peers due to shared interests.
 - (b) Greater academic achievements due to logical thinking skills gained from gambling.
 - (c) Increased peer pressure and strained familial relationships.
 - (d) Enhanced understanding of economic principles.
 - (e) Development of leadership skills through betting platforms.
59. What financial risk is associated with youth engaging in betting, based on the passage?
- (a) Long-term investment planning difficulties.
 - (b) Lack of interest in career development.
 - (c) Falling into a cycle of chasing losses due to impulsive betting.
 - (d) Accumulation of wealth through regular gambling.
 - (e) Excessive saving habits that affect spending.
60. What is suggested as a possible solution to mitigate the negative effects of betting app advertisements on youth?
- (a) Promoting gambling as a harmless activity.
 - (b) Introducing stricter regulations on online advertising.
 - (c) Encouraging families to monitor all social media usage.
 - (d) Focusing on the benefits of gambling education.
 - (e) Eliminating all forms of digital advertisements.

Directions (61-62): Read the following passage and answer the given questions.

China's Gross Domestic Product (GDP) is one of the largest in the world, reflecting its position as a global economic powerhouse. Currently, China has the second-largest GDP, valued at over \$17 trillion,

trailing only the United States. This growth has been driven by industrial development, strong exports, and increasing technological innovation.

For many years, China's economy was centered on manufacturing, earning it the title "the factory of the world." However, recent trends show a shift towards services and domestic consumption, with e-commerce and technology playing key roles in driving growth. Companies like Alibaba and Huawei highlight this transition.

Despite its rapid expansion, China faces challenges such as an ageing population, environmental concerns, and trade tensions with other nations. To address these issues, the government is focusing on green energy, innovation, and infrastructure development.

China's GDP growth remains steady, ensuring its influence on the global economy continues to grow.

61. What can be inferred about the shift in China's economy in recent years?
- China has completely abandoned manufacturing for services.
 - Manufacturing remains important, but services and domestic consumption are becoming significant drivers of growth.
 - The services sector has replaced all other sectors in contributing to GDP growth.
 - Exports have stopped contributing to China's GDP.
 - China is reducing its focus on technology and e-commerce.
62. What is a likely reason for China's focus on green energy and innovation?
- To replace manufacturing with agriculture as the primary sector.
 - To address challenges like environmental concerns and slowing growth.
 - To increase its reliance on foreign investments.
 - To decrease the value of its GDP intentionally.
 - To reduce the need for domestic consumption.

Directions (63-67): In the following passage, some of the words are missing. Find the suitable word from the options given below that make the paragraph meaningful.

Climate negotiations have a tendency to get entangled more in words than in enabling action. A devastating year of heat waves, droughts, lost crops, extreme floods made it impossible to ignore climate-related damages to lives and livelihoods. The latest round of _____ (A), COP27 in Egypt began with little hope and ended two days beyond schedule with a faint glimmer. All 197 Parties to the UN Framework Convention on Climate Change (UNFCCC) agreed to enable financing for loss and damage to those that need it the most. But the goal of keeping temperature rise to levels that could _____ (B) loss and damage continued to slip away. Loss and damage (or L&D) refers to the adverse impacts that _____ (C) communities and countries face as a result of a changing climate, including the increased incidents and intensity of natural disasters and extreme weather as well as slow-onset temperature increase, sea-level rise, or desertification. Rich countries had resisted L&D payments for years, worried that their historical emissions would impose large calls for compensation. Under pressure, they could no longer duck their _____ (D). The COP27 decision includes the development of a Transition Committee dedicated to L&D, with equal representation across rich and poor countries. In order to operationalise the funding arrangements, this committee has been tasked with configuring institutional _____ (E), identifying and expanding sources of funding, and coordinating with existing funding arrangements — all by COP28 in the UAE next year. The sticking points of negotiations, thus, remain. India can speak — and act — on behalf of the Global South, to ensure that the new L&D financing facility delivers. A global index, based on climatic risks and compounding impacts, would transparently—and scientifically—establish the sheer size of the problem that the L&D facility will have to address.

63. Which of the following options is suitable for blank (A)?
- Dialogue
 - Discussions
 - Examination

- Only (I)
- Only (II)
- Only (III)
- Both (I) and (II)
- Both (I) and (III)

64. Which of the following options is suitable for blank (B)?
- Exacerbate
 - Exaggerate
 - Minimize
- Only (I)
 - Only (II)
 - Only (III)
 - Both (I) and (II)
 - Both (I) and (III)
65. Which of the following options is suitable for blank (C)?
- Vulnerable
 - Distinctive
 - Rapport
- Only (I)
 - Only (II)
 - Only (III)
 - Both (I) and (II)
 - Both (I) and (III)
66. Which of the following options is suitable for blank (D)?
- Impediment
 - Onus
 - Responsibility
- Only (I)
 - Only (II)
 - Only (III)
 - Both (I) and (II)
 - Both (I) and (III)
67. Which of the following options is suitable for blank (E)?
- Advancements
 - Arrangements
 - Channelization
- Only (I)
 - Only (II)
 - Only (III)
 - Both (I) and (II)
 - Both (I) and (III)

Directions (68-71): The following questions have two statements with connectors as alternatives. You must choose the one that will link both statements and make them grammatically and contextually accurate.

68. I. Sharan and Ram have incredible voices
II. They will go far in their music career.
- Similarly
 - Otherwise
 - Undoubtedly
- Only A
 - Only B
 - Both A and C
 - Both A and B
 - Both B and A
69. A. Any person contravening this order shall be punished.
B. Section 188 of the Indian Penal Code (IPC) and appropriate sections of the Epidemic Diseases Act and National Disaster Management Act.
- Even
 - With
 - Under
- Only (ii)
 - Both (i) and (iii)
 - Only (i)
 - All (i), (ii) and (iii)
 - Only (iii)
70. A. Originally called Abhijit Dhananjay Sarang, Kalicharan reportedly changed his name.
B. He met with an accident while in his teens that badly twisted his leg but left him with only a hairline fracture.
- While
 - After
 - Since
- Only (i)
 - Only (ii)
 - Only (iii)
 - Both (i) and (ii)
 - All (i), (ii) and (iii)
71. A. In the clip, a group of tourists is seen sitting inside a safari vehicle, with wired mesh covering the windows.
B. As the car stands on the road for a better view, the big cat attempts to bite off the rear bumper of the black car.
- However
 - Though
 - Meanwhile
- Only (iii)
 - Both (i) and (iii)
 - Only (ii)
 - Only (i)
 - None of these

Directions (72-75): In the following question, a sentence is given in which the last part is omitted. You must choose the most suitable alternative among the five that should fill the omitted part making the sentence grammatically correct and contextually meaningful. If none of the given alternatives are appropriate to fill the blank, choose option (e) i.e. "none of these" as your answer choice.

72. Twitter does not want users to migrate to other platforms through links shared via tweets or retweets to ensure users spend more time on the platform which in turn _____.
 (a) short-form content while using recommendations to drive discovery across the platform.
 (b) which is also actively looking to increase its paid subscriber count.
 (c) it would no longer allow users to promote their presence on other social media platforms.
 (d) boosts its position to increase advertising revenue.
 (e) None of these
73. Financial inclusion, a cornerstone of equitable economic growth, aims to ensure that individuals and communities _____.
 (a) savings, credit, insurance, and investment opportunities.
 (b) financial systems that were previously inaccessible.
 (c) have access to a wide range of financial services and resources.
 (d) limits their ability to invest in education, healthcare, and entrepreneurship.
 (e) None of these
74. Climate change remains a critical issue on the global agenda as the world _____.
 (a) address rising global temperatures and extreme weather events.
 (b) grapples with its far-reaching implications.
 (c) focus on renewable energy, carbon neutrality, and nature-based solutions.
 (d) limit the worst impacts of climate change and protect the planet's ecosystems.
 (e) None of these
75. Investment and funding are integral components of the financial landscape that significantly impact _____.
 (a) with the expectation of generating returns over a specific period.
 (b) entrepreneurial endeavors that exhibit promising growth potential.
 (c) to facilitate diverse initiatives, projects, or business ventures.
 (d) economic development, technological innovation, and the realization of various projects.
 (e) None of these

76. **Direction:** In each of the questions given below, a passage has been given which is either situation based or describes a scenario. Below each passage, a question has been given which is framed to make you infer something out of it. Mark out the correct inference as your answer.

The world will be transformed. By 2050, we will be driving electric cars and flying in aircraft running on synthetic fuels produced through solar and wind energy. New energy-efficient technologies, most likely harnessing artificial intelligence, will dominate nearly all human activities from farming to heavy industry. The fossil fuel industry will be in the final stages of a terminal decline. Nuclear fusion and other new energy sources may have become widespread. Perhaps our planet will even be orbited by massive solar arrays capturing cosmic energy from sunlight and generating seemingly endless energy for all our needs.

Which of the following is true about the role of AI in future clean energy goals?

- (a) AI will dominate the harnessing of new clean energy.
 (b) New clean energy will be harnessed with some help from AI.
 (c) AI will overtake nuclear fusion as a source of clean energy harnessing processes.
 (d) Both A and B
 (e) Both B and C

Directions (77-78): the question below contains five scattered segments of a sentence. Indicate the sequence which correctly assembles the segments and completes the sentence.

77. A. lakes, and coastal areas in reducing physical exposure to
 B. natural hazards such as floods, landslides
 C. or avalanches by serving as buffers
 D. there are many experiences around the world which point to the potential
 E. benefits of protecting and preserving ecosystems such as wetlands, forests
 (a) DECBA (b) ABCED (c) CABDE
 (d) ABDEC (e) DEABC
78. A. Even during the lockdown, people cooped up in homes with less than
 B. usual activity were advised by psychologists to take time off
 C. and relax in places close to home where one
 D. for mental and physical well being.
 E. feels less stressed, as it is important
 (a) DECBA (b) ABCED (c) CABDE
 (d) ABDEC (e) DEABC

Directions (79-80): Below given is a passage with the 5 words in BOLD. Answer the questions following the passage.

Rahul and Ananya were excited about the new **zipline** (1) park opening nearby. They had heard about thrilling activities like the **adventure** (2) and rock climbing, which sparked their curiosity. While Ananya was eager to try everything, Rahul felt slightly nervous about **safety** (3). However, Ananya **reassured** (4) him by mentioning the park's focus on **heights** (5) and the presence of trained staff. They decided to visit early in the morning to avoid the long **queues** (6), with Ananya taking charge of booking the tickets online. Both friends looked forward to a day filled with fun and excitement.

79. Find the correct combination of the words that replace each other so as to make the given paragraph contextually and grammatically correct. If the sentence is correct as it is then select option (e) as your choice.
 (a) 1-4 and 2-3 (b) 2-1 and 3-5
 (c) 2-3 and 4-5 (d) 2-4 and 1-5
 (e) No interchange required
80. What is the synonym of the word adventure?
 (a) Danger (b) Exploration
 (c) Routine (d) Obstacle
 (e) Boredom

DATA ANALYSIS AND INTERPRETATION

81. Two series I and II given below each have one wrong number P and Q respectively.

Series I: 5, 20, 40, 67, 105, 140, 224

Series II: 13, 20, 30, 98, 119, 334

Find the difference between P and Q ?

- (a) 24 (b) 36 (c) 38
 (d) 42 (e) None of these

Directions (82-83): Read the following passage and answer the given questions.

Two algebraic expressions are given as (I) $6x^2 + 29x + P$ and (II) $3x + 1$. (I) is divided by (II) leaving $(2x + 9)$ as quotient and 19 as remainder.

82. Find the value of P.
 (a) 25 (b) 26 (c) 27
 (d) 28 (e) None of these
83. What is the value of largest root of equation $6x^2 + 29x + P = 0$?
 (a) $-1\frac{1}{3}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$
 (d) $\frac{1}{4}$ (e) None of these

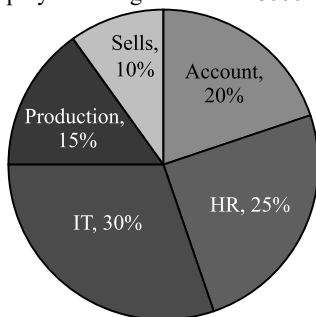
84. Puneet buys 240 articles and labels each with the same marked price. He sells 160 articles initially at 20% discount on the labelled price. Then he sells rest articles at an additional discount of 25% on the discounted price. Thus, he gets total of ₹Z, and makes profit of 10%. When he offered no discount find its profit %.
- (a) 50% (b) 25% (c) 20%
(d) 33% (e) None of these

85. The container X contains 140 liters pure milk. $\frac{250}{7}$ % of the milk is taken out from the container and 30 liters of water is added in it and then another 40% of the mixture is taken out. If another container Y has 80 liters of mixture in which 62.5% is milk and remaining soda, then find the final quantity of water in container X is what percent of quantity of soda in container Y ?
- (a) 65% (b) 60% (c) 75%
(d) 25% (e) None of these

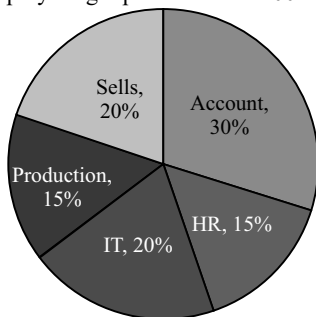
Directions (86-90): Read the following passage and answer the given questions.

Study the following given pie - charts carefully and answer the questions given below. The first pie chart represents the breakup of the number of employees working in different departments of an organization and the second pie chart represents the distribution of promotion of employees.

Total number of employees in organization = 6000



Total number of employees got promotion = 1200



86. If the number of female employees who were promoted in HR department were 120 and the ratio of male and female employees in HR department is 7 : 5, then the ratio of the total number of male employees in the HR department to the number of male employees promoted in the same department was:
- (a) 105 : 8 (b) 112 : 15 (c) 145 : 11
(d) 175 : 12 (e) None of these.
87. If the ratio of number of male employees to the female employees in the production department is 2 : 3 and the ratio of number of male employees to female employees who get promotions in the same department is 4 : 5, then what percent of the female employees get promotions in the production department?
- (a) 18.52% (b) 21.50% (c) 22.50%
(d) 25% (e) 33.33%

88. The difference between the percentage of the number of employees, who get promotions from the HR department and IT department is:
- (a) 0.50% (b) 0.75% (c) 1.25%
(d) 1.33% (e) 1.60%
89. The total number of employees promoted in the Account, IT, and Sells department together is what percent of the total number of employees in these departments?
- (a) 18.33% (b) 23.33 % (c) 22.02 %
(d) 33.13 % (e) None of these
90. What is the difference between the average number of employees who were promoted in the IT, Production, and Sells departments and the average of number of total employees in these three departments?
- (a) 540 (b) 740 (c) 880
(d) 950 (e) 1250
91. A bag contains some balls of three different colours. Probability of drawing a blue colour ball is $\left(\frac{5}{24}\right)$ and the ratio of red to green colour balls is 9 : 10. What is the total number of balls in the bag?
- I. Average of total number of balls of all the three colours is 8.
II. Two balls are drawn randomly from the bag and probability that both the balls are green is $\frac{15}{92}$.
III. One ball is drawn at random from the bag and the probability that the ball is either blue or red is $\left(\frac{7}{12}\right)$.
- (a) Either I alone or III alone is sufficient.
(b) Either I alone or II alone is sufficient.
(c) Either II alone or III alone is sufficient.
(d) Any one of them alone is sufficient.
(e) Any two of them together are sufficient.
92. Question below is followed by three statements I, II and III. You have to determine whether the data given in the statement is sufficient for answering the question.
- y litres mixture of milk and water contains milk and water in the ratio $a : b$ respectively. The milkman sold z litres of the mixture and added 10 litres of pure milk and 8 litres of water to the remaining mixture such that the ratio of milk and water in the final mixture became 5 : 4 respectively. Find the respective ratio of amount of milk in the initial mixture and amount of milk in the final mixture.
- Statement I:** A 54 litres mixture of syrup and water contains syrup and water in the ratio $a : b$ respectively. If 10 litres of syrup and 6 litres of water is added to the mixture, the ratio of syrup and water becomes 4 : 3 respectively.
- Statement II:** A 60 litres mixture of milk and water contains milk and water in the ratio 7 : 5 respectively. If z litres of the mixture is replaced with water, the ratio of milk and water in the mixture becomes 7 : 23 respectively.
- Statement III:** $y = 3z$
- (a) Only I and II together (b) Either I or II and III together
(c) Any two of the three (d) All I, II and III together
(e) Cannot be answered even after combining all the three statements
93. Time taken by the boat in river A to cover $(D + 120)$ km in downstream is 25% more than the time taken by the same boat to cover $(D - 80)$ km in upstream in river B. The speed of the stream in river B is 20% less than the speed of stream in river A. Ratio of the speed of the boat to the speed of the stream in river A is 9 : 5. Find the value of $(D + 40)$ if the speed of the boat is constant.
- (a) 160 (b) 200 (c) 240
(d) 280 (e) None of these

94. Ratio of the height of the cone and cylinder is 3 : 2. Ratio of the radius of the cone and cylinder is 3 : 4. Ratio of the volume of the cone and cylinder is $m : n$. The length of the cuboid is $(m + 11)$ and the breadth of the cuboid is $(n - 11)$. The total volume of the cuboid is 11340. Find the possible value of the height of the cuboid?

- I. $3m$
 II. $\frac{n}{4}$
 III. $n - m - 15$
 (a) Only I (b) Only II (c) Only III
 (d) Only I and II (e) Only I and III

Directions (95-98): Read the following information carefully and answer the given questions.

A shopkeeper sold five articles: P, Q, R, S and T. Further partial information is given in table below.

Article	Cost price	% Profit	Marked price
P	—	40%	$(10.8b + 3m)$
Q	—	—	$(12a - n)$
R	—	20%	—
S	$(5b - m - n - 48)$	—	—
T	$\left(4a + \frac{m}{2}\right)$	—	$\left(13a - \frac{n}{2} + 6\right)$

Note:

- (i) Value of 'n' is 4 times of larger root of given equation.
 $K^2 - 32P + 252 = 0$
- (ii) Article T is marked up ₹1280 more than its cost price and the difference between marked price and cost price of article T is 28% more than profit earned on article S.
- (iii) Discount given on article Q is 16.66%, and selling price of S is ₹ $(5m + 7n + 4a + 296)$.
- (iv) Value of 'm' is twice of missing value in the given sequence.
 140, 136, 161, (?), 329, - 200
95. If cost price of P is 87.5% more than that of T, and P is sold after discount of $(2L - 210)\%$, then which of the following is/are possible value of L?
- I. 1.5 m
 II. $(b - n - 8)$
 III. $2.4(b - a)$
 IV. 240
 (a) I and III only (b) II and IV only
 (c) I, II, and III only (d) I, II, III, and IV
 (e) None of these
96. If marked price of T is ₹80 less than cost price of R, which is sold after discount of ₹400 or $\left(\frac{5L}{14}\right)\%$. Find the value of L.
- (a) $\frac{100}{7}$ (b) 40 (c) 250
 (d) $\frac{2}{5}$ (e) None of these
97. If profit% given on article Q is thrice as that of discount %, then find difference between cost price of Q and S.
 (a) 160 (b) 80 (c) 120
 (d) 240 (e) None of these
98. If the discount given on article T is $(b - a)\%$ and sold at a profit of L%, while the marked price of S is ₹400 less than that of P, and S sold after giving a discount of D%. Find $(L\% - D\%)$.

- (a) 30% (b) 20% (c) 25%
 (d) 40% (e) None of these

Directions (99-101): Read the following passage and answer the given questions.

There are five pouches (A, B, C, D, and E) which contain candies of four flavors (Orange, Pineapple, Caramel, and Coffee). Below table shows the number of candies of four different flavors in five pouches:

	Orange candy	Pineapple candy	Caramel candy	Coffee candy
Pouch A	12	---	---	---
Pouch B	---	24	---	10
Pouch C	---	---	16	---
Pouch D	---	---	10	---
Pouch E	---	10	---	---

Some information is given below:

- When one candy is picked at random from pouch A, then the probability of getting one coffee candy is $\frac{1}{3}$ and when one candy is picked at random from pouch C, then the probability of getting neither caramel nor coffee is $\frac{1}{2}$.
 - In pouch E, number of orange candies is same as number of caramel candies. Total number of caramel candies in all five pouches together is 69.
 - The number of pineapple candies in pouches C and D are equal and when two candies are picked at random from pouch A, then the probability of getting both being pineapple candies is $\frac{1}{22}$.
 - When one candy is picked at random from pouch B, then the probability of getting either orange candy or caramel candy is $\frac{25}{42}$. Total number of caramel candies and coffee candies are same in all five pouches together. Number of caramel candies in pouch D is 2 more than that in pouch A.
 - When one candy is picked at random from pouch C, then the probability that the candy picked is not caramel flavor is $\frac{4}{5}$. When two candies are picked at random from pouch D, then the probability that none of the candies is orange flavor is $\frac{17}{35}$.
 - When one candy is picked at random from pouch E, then the probability of getting one orange candy is $\frac{1}{6}$. When one candy is picked at random from pouch D, then probability of getting coffee candy is $\frac{1}{5}$. When one candy is picked at random from pouch E, then the probability of getting one pineapple candy is $\frac{1}{3}$.
99. The question below contains a statement followed by Quantity I, Quantity II and Quantity III. Find all quantities to find the relationship among them.
- If 5 orange candies from pouch A are shifted to pouch C and 2 caramel candies from pouch D are shifted to pouch A.
- Quantity I:** When two candies are picked at random from pouch D, then what is the probability that the both candies are caramel flavor?
- Quantity II:** When three candies are picked at random from pouch A, then what is the probability that exactly two candies are pineapple flavor?
- Quantity III:** When one candy is picked at random from pouch C, then what is the probability that the picked candy is orange flavor?

- (a) Quantity I > Quantity II < Quantity III
 (b) Quantity I > Quantity II > Quantity III
 (c) Quantity I < Quantity II > Quantity III
 (d) Quantity I < Quantity II < Quantity III
 (e) Quantity I = Quantity II < Quantity III
100. In pouch B, 25% of candies are rotten in which one-third is caramel flavor. If two candies are picked at random from pouch B, then what is the probability of getting one rotten caramel candy and another good candy?
- (a) $\frac{24}{167}$ (b) $\frac{21}{166}$ (c) $\frac{23}{165}$
 (d) $\frac{21}{164}$ (e) $\frac{19}{166}$
101. If 'K' pineapple candies are taken out from pouch D and put into pouch E, and then '2K' caramel candies and 'K' coffee candies are taken out from pouch E and put into pouch D. One candy is picked at random from pouch D, then the probability of getting orange flavor candy is $\frac{1}{4}$. What is the value of 'K'?
- (a) 30 (b) 7 (c) 10
 (d) 2 (e) 5
102. Time taken by P, Q, and R together to complete the job is 37.5% less than time taken by Q and R together to complete the job, while P and Q together can do the job in Z days, and R alone can complete it in Y days. Share of R is ₹8000 out of total share of P, Q, and R together (₹16000). If P, Q and R together can complete the job in (Z - K) days, which of the following is/are definitely true.
- I. $\frac{Y}{2} = K$
 II. If time taken by P, Q and R together to complete the job is 15 day less than that of P and Q together, then time taken by S to complete the job is 20 days, if R is half efficient that as that of S.
 III. Z = Y
- (a) III only (b) I and III only
 (c) II and I only (d) II and III only
 (e) None of these
103. Car P and Q started from point A and point B respectively running towards each other. Car P running at 4Z km/h and meets Q after 10 hours. After reaching point A, driver of car Q realized that car R (running towards point B) is $\left(\frac{3Y}{5} - 360\right)$ km ahead of car Q. So, car Q started chasing car R and overtakes it in 6 hours. Find speed of car P, if distance between point A and point B is Y km and speed of car R is 2Z km/h.
- (a) 20 km/hr (b) 40 km/hr
 (c) 60 km/hr (d) 80 km/hr
 (e) None of these
104. If value of M = 30, then for which of the following series value of M not satisfied properly?
- I. 18, M, 50, 97, 208, 444 II. -50, -49, -45, -35, -16, $\frac{M}{2}$
 III. 8, 5, 6.5, M, 110, 1319
- (a) Only I (b) Both II and III
 (c) Only III (d) Both I and II
 (e) None of these
105. In the question, three series I, II and III are given. Find the value of x, y and z to establish the correct relation among them and choose the correct option.
- (i) 12, 24, 8, x, 6.4, 38.4
 (ii) 11, 20, y, 50, 71, 96
 (iii) 3, 4, 6, 12, z, 156
- (a) $x = y = z$ (b) $x < y = z$ (c) $x = y < z$
 (d) $x < y < z$ (e) $x > y > z$

106. Given below three series I, II & III given and each series has a wrong number. The number that should come in place of the wrong number in series I, II & III is P, Q & R respectively. Find the value of P, Q & R and then find which of the following statement is true.
- 244, 380, 476, 536, 568, 580, 580
 160, 200, 480, 1200, 3600, 12600, 50400
 15, 20, 36, 65, 111, 186, 264
- (a) $P = Q < R$ (b) $P = Q > R$
 (c) $P > Q > R$ (d) $P < Q < R$
 (e) None of these
107. Calculate the value of (A) and (B) and answer the questions given below:
- (A), (B), 102, 104, 107, 113, 128, 173
- If M is 150% more than (A), then find which of the following number is nearest square root of the difference between M and (B).
- (a) 16 (b) 12 (c) 9
 (d) 18 (e) None of these
108. **Direction:** In the following question, two quantities I and II are given. Compare quantity I and quantity II on its basis. (Only quantity is to be considered)
- Two trains A and B are running in the same direction and they cross a 75 m long platform in 25 seconds and 15 seconds respectively. Speed of train B is 25.2 km/h more than that of train A and the length of train A is 45 m more than that of train B.
- Quantity I:** If both the trains were running in opposite directions, then in what time would they cross each other?
Quantity II: If the speed of train B were 3 m/s more than its original speed, then at what time would train B cross a 95 m long tunnel?
- (a) Quantity I > Quantity II (b) Quantity I < Quantity II
 (c) Quantity I ≥ Quantity II (d) Quantity I ≤ Quantity II
 (e) Quantity I = Quantity II or relation can't be established
109. **Direction:** In the following question, two quantities I and II are given. Compare quantity I and quantity II on its basis. (Only quantity is to be considered)
- Train A crosses train B in 13 seconds while running in the opposite direction and train C crosses train A in 100 seconds while running in the same direction. Average of speeds of trains A and B is 15 m/s and the speed of train C is 4 m/s more than that of train A. Train A can cross a 140 m long platform in 20 seconds and the speed of train B is 14 m/s.
- Quantity I:** If speed of train C were 95% of its original speed, then in what time train C would cross train B while running in the same direction?
Quantity II: If a series is formed, whose 1st term is the length of train A, 2nd term is (1st term - 2²), 3rd term is (2nd term - 3²), 4th term is (3rd term - 5²) and 5th term is (4th term - 7²), then what is the 5th term of the series?
- (a) Quantity I > Quantity II (b) Quantity I < Quantity II
 (c) Quantity I ≥ Quantity II (d) Quantity I ≤ Quantity II
 (e) Quantity I = Quantity II or relation can't be established
110. **Direction:** In the following question, three quantities I, II and III are given. Compare quantity I, quantity II and quantity III on its basis. (Only quantity is to be considered)
- Two equations (a) and (b) are given below:
- (a) $3x - 4y = 2$
 (b) $x - 2y = 8$
- Quantity I:** Find the value of 'x' by solving both the equations?
Quantity II: Find the value of 'y' by solving both the equations?
Quantity III: An article is marked up by ₹250 and sold by giving a discount of ₹200 on marked price. If the marked up percent is P%, discount given is Q% and marked price of the article is ₹1000, then find the value of (Q - P)?

- (a) Quantity I > Quantity III > Quantity II
 (b) Quantity II > Quantity III > Quantity I
 (c) Quantity III < Quantity II < Quantity I
 (d) Quantity II < Quantity I < Quantity III
 (e) Quantity I = Quantity II < Quantity III

111. Directions: In the following question, two quantities I and II are given. Compare quantity I and quantity II on its basis. (Only quantity is to be considered).

Two series A and B are given below with one wrong term in series.

A: 3, 8, 24, 45, 120, 168

B: 28, 42, 56, 98, 154, 182

Quantity I: Average of wrong terms of both the series.

Quantity II: Difference between the next term of series A and the next term of series B.

- (a) Quantity I > Quantity II (b) Quantity I < Quantity II
 (c) Quantity I ≥ Quantity II (d) Quantity I ≤ Quantity II
 (e) Quantity I = Quantity II or No relation

Directions (112-115): Read the following information carefully and answer the given questions.

Some recruitment is done by three companies, i.e., P, Q and R. The number of vacancies in company P, Q, and R are 6, 12 and 2 respectively. The number of candidates, who got interview call for per vacancy in company P, Q and R are 60, 6x and 4x respectively. The total number of candidates applied in company R is 25% of the total number of candidates, applied in the company Q and three fifth of the total number of candidates applied in the company P. The difference between the number of candidates, who got interview calls in the company Q and R is 320.

- 112.** What is the total number of candidates applied in company Q?
 (a) 450 (b) 600 (c) 300
 (d) Can't be determined (e) None of these
- 113.** The number of candidates applied in company R is what percentage of the number of candidates applied in company P?
 (a) 75% (b) 50% (c) 60%
 (d) 80% (e) None of these
- 114.** In company P, out of the total number of candidates who got interview call 40% are female and company P allotted its posts to male and female in the ratio of 2 : 1 respectively, then find the number of females who did not get the job in company P.
 (a) 142 (b) 90 (c) 62
 (d) 150 (e) None of these
- 115.** If the difference between the total number of candidates applied in company P and company R is 160, then find the number of candidates, who did not get interview call from company Q.
 (a) 900 (b) 960 (c) 1050
 (d) 600 (e) None of these

GENERAL, ECONOMY/BANKING AWARENESS

- 116.** What is the maximum limit for UPI123 Pay and UPI Lite Wallet transactions respectively?
 (a) ₹15,000 for UPI123 Pay, ₹7,000 for UPI Lite Wallet
 (b) ₹10,000 for UPI123 Pay, ₹5,000 for UPI Lite Wallet
 (c) ₹12,000 for UPI123 Pay, ₹6,000 for UPI Lite Wallet
 (d) ₹5,000 for UPI123 Pay, ₹10,000 for UPI Lite Wallet
 (e) ₹8,000 for UPI123 Pay, ₹4,000 for UPI Lite Wallet
- 117.** What is the implementation period for the National Green Hydrogen Mission (NHM)?
 (a) 2023-2025 (b) 2023-2030
 (c) 2024-2032 (d) 2025-2030
 (e) 2024-2035

- 118.** Which company manufactures the Vande Bharat train in India?
 (a) Hindustan Aeronautics Limited (HAL)
 (b) Tata Motors
 (c) Integral Coach Factory (ICF)
 (d) Bharat Heavy Electricals Limited (BHEL)
 (e) Larsen & Toubro (L&T)
- 119.** What is the interest rate for the Sukanya Samridhi Scheme for the quarter from October 1, 2024, to December 31, 2024?
 (a) 7.1% (b) 7.5% (c) 8.2%
 (d) 7.7% (e) 4.0%
- 120.** Where has HDFC Bank recently opened its first branch as part of its international expansion strategy?
 (a) Dubai (b) London (c) New York
 (d) Singapore (e) Tokyo
- 121.** What is the Return on Assets (RoA) for Scheduled Commercial Banks (SCBs) as of March 2024, according to the 29th issue of the Financial Stability Report (FSR) released by the Reserve Bank of India (RBI)?
 (a) 1.0 percent (b) 1.3 percent
 (c) 1.5 percent (d) 2.0 percent
 (e) 2.5 percent
- 122.** Which public sector bank recorded the highest profit in FY 2024?
 (a) State Bank of India (b) Bank of Maharashtra
 (c) Bank of Baroda (d) Union Bank of India
 (e) Indian Bank
- 123.** What is India's rank in the 2024 Global Nature Conservation Index (NCI) developed by the Goldman Sonnenfeldt School of Sustainability and Climate Change at Ben-Gurion University and BioDB.com, a biodiversity database?
 (a) 156th (b) 167th (c) 176th
 (d) 178th (e) 180th
- 124.** Who is the present Union Minister of Agriculture and Rural Development?
 (a) Narendra Singh Tomar (b) Giriraj Singh
 (c) Shivraj Singh Chauhan (d) Manohar Lal Khattar
 (e) Parshottam Rupala
- 125.** In November 2024, the Ministry of Railways planned to cover over 66,000 route km (rkm) out of its total network with the indigenous Automatic Train Protection (ATP) system named Kavach, by _____.
 (a) 2025 (b) 2028 (c) 2027
 (d) 2030 (e) 2035
- 126.** What is the Paid-up- Capital of the Reserve Bank of India?
 (a) 5 crore (b) 10 crore (c) 15 crore
 (d) 20 crore (e) 50 crore
- 127.** Under Basel III, what is the minimum Common Equity Tier 1 (CET1) capital requirement for banks in India?
 (a) 4.5% (b) 6% (c) 8%
 (d) 10.5% (e) 12%
- 128.** Where will the Hockey Women's Asian Champions Trophy 2024 be held?
 (a) Ranchi, Jharkhand (b) New Delhi, India
 (c) Rajgir, Bihar (d) Bangalore, Karnataka
 (e) Mumbai, Maharashtra
- 129.** The maximum limit of SLR in India is _____.
 (a) 40% (b) 20% (c) 25%
 (d) 36% (e) 50%
- 130.** In India, what percentage is set as the Priority Sector Lending (PSL) goal for Small Finance Banks (SFBs)?
 (a) 50% (b) 60% (c) 65%
 (d) 75% (e) 80%
- 131.** Which of the following banks has recently unveiled a research study named 'Of the Women, By the Women, For the NATION'?

highlighting the evolving dynamics of Self-Help Groups (SHGs) in India and delves into the patterns of credit utilisation and digital behaviours among SHGs, their members, and the emerging group known as ‘Lakhpatti Didis’?

- (a) State Bank of India (b) Indian Bank
(c) Union Bank of India (d) Canara Bank
(e) Bank of Baroda

132. Railway Minister Ashwini Vaishnaw reviewed the trial of the Automatic Train Protection System, ‘Kavach’, and oversaw the upgrades made to the system. In how many different emergency situations did the Kavach system function as expected during the trial?

- (a) 3 (b) 5 (c) 7
(d) 10 (e) 4

133. What action has the RBI instructed the four NBFCs—Asirvad Micro Finance Ltd, Arohan Financial Services Ltd, DMI Finance Pvt Ltd, and Navi Finserv Ltd—to take as part of the imposed restrictions?

- (a) Halt loan recovery operations
(b) Cease and desist from sanctioning and disbursing loans
(c) Shut down their operations permanently
(d) Reduce interest rates on existing loans
(e) Merge with larger NBFCs

134. What is the name of the new large language model (LLM) introduced by Meta at Meta Connect 2024?

- (a) Llama 3.0 (b) Llama 3.2
(c) Llama Vision (d) Llama AI
(e) Meta Vision Llama

135. What is the new position limit for trading members in index futures and options as announced by SEBI?

- (a) ₹5,000 crore or 10% of total open interest, whichever is higher
(b) ₹7,500 crore or 15% of total open interest, whichever is higher
(c) ₹6,000 crore or 12% of total open interest, whichever is higher
(d) ₹7,000 crore or 14% of total open interest, whichever is higher
(e) ₹7,500 crore or 20% of total open interest, whichever is higher

136. India and which country signed a Memorandum of Implementation (MoI) for the co-development of the UNICORN (Unified Complex Radio Antenna) mast designed to enhance the stealth characteristics of warships and will be fitted on Indian Naval warships?

- (a) Japan (b) Germany (c) Australia
(d) Spain (e) France

137. What is the loan amount provided by the International Finance Corporation (IFC) to Axis Bank for blue finance and green projects in India?

- (a) \$300 million (b) \$400 million
(c) \$500 million (d) \$600 million
(e) \$700 million

138. What is the total outlay approved by the Union Cabinet for the seven schemes aimed at improving farmers’ lives?

- (a) ₹10,500 crore (b) ₹12,000 crore
(c) ₹14,235.30 crore (d) ₹16,500 crore
(e) ₹20,000 crore

139. What is the prize money for the Booker Prize 2024?

- (a) £40,000 (b) £45,000 (c) £50,000
(d) £60,000 (e) £75,000

140. The India Space Research Organization’s Pushpak mission is associated with _____.

- (a) Developing a manned lunar exploration program
(b) Establishing a space station
(c) Developing a reusable launch vehicle (RLV)
(d) Building interplanetary spacecraft
(e) Conducting Mars and Venus exploration

141. As per the guidelines of the Reserve Bank of India, what is the ongoing minimum percentage for the Capital to Risk-weighted Assets Ratio (CRAR) that Cooperative Banks (UCBs) must uphold?

- (a) 3 percent (b) 5 percent
(c) 6 percent (d) 7 percent
(e) 9 percent

142. Gross Merchandise Value (GMV) of the Government e-Marketplace (GeM) during the financial year FY24 was ₹4 trillion. What percentage of the total order value on GeM during FY24 was contributed by Micro and Small Enterprise (MSE) sellers?

- (a) 25.63% (b) 35.71% (c) 45.82%
(d) 49.37% (e) 55.24%

143. Which among the following is not one of the promoters of the National Centre for Financial Education (NCFE), created for the promotion of financial education across India as per the National Strategy of Financial Education?

- (a) Reserve Bank of India (RBI)
(b) Securities and Exchange Board of India (SEBI)
(c) Insurance Regulatory and Development Authority of India (IRDAI)
(d) Pension Fund Regulatory and Development Authority (PFRDA)
(e) Employees’ Provident Fund Organization.

144. What company has received approval for its merger with North East Small Finance Bank (NESFB)?

- (a) Razorpay (b) Paytm
(c) Slice (d) PhonePe
(e) Zomato

145. Which of the following is NOT one of the five pillars of the Reserve Bank of India’s (RBI) Payments Vision 2025?

- (a) Integrity (b) Innovation
(c) Internationalization (d) Incorporation
(e) Institutionalisation

146. What is the expected launch timeline for India’s Venus Orbiter Mission (VOM) unofficially referred to as “Shukrayaan”?

- (a) March 2024 (b) March 2025
(c) March 2028 (d) June 2029
(e) September 2030

147. As per the Reserve Bank of India (RBI) regulations, the minimum Net Owned Fund (NOF) requirement for NBFC-Micro Finance Institutions (NBFC-MFIs) is:

- (a) ₹2 Crore (b) ₹3 Crore (c) ₹5 Crore
(d) ₹7 Crore (e) ₹10 Crore

148. Who is the director of the film “Laapataa Ladies”, India’s official entry for the Oscars?

- (a) Kiran Rao (b) Amir Khan
(c) Karan Johar (d) Rajkumar Hirani
(e) Frahan Akhtar

149. Which has become the first foreign university to receive approval of the International Financial Services Centres Authority (IFSCA) to set up an International Branch Campus (IBC) in GIFT-IFSC, GIFT City, Gujarat?

- (a) Deakin University, Australia
(b) National University of Singapore (NUS)
(c) Tokyo University, Japan
(d) RMIT University
(e) Bandung Institute of Technology, Indonesia

150. Which state is the top performer in terms of the number of micro, small, and medium enterprises (MSMEs) in India as of March 2024?

- (a) Uttar Pradesh (b) Maharashtra
(c) Tamil Nadu (d) Gujarat
(e) Rajasthan

151. Consider the following statements regarding budget allocation for Renewable Energy:

- I. The budget allocated ₹19,100 crore for the Ministry of New and Renewable Energy.
- II. The allocation for the solar energy sector was ₹10,000 crore, reflecting an increase of 110% from ₹4,757 crore in the previous fiscal year (2023-24).
- III. The PM Surya Ghar Muft Bijli Yojana (PMSGMBY) was announced to boost RTS penetration in India, with a total budget of ₹75,000 crore until 2027.

Which of the statements given above is/are correct?

- (a) I and II only (b) I and III only
(c) II and III only (d) All I, II and III
(e) I only

152. 'Incoterms', the short form of International Commercial Terms, is a registered trademark of the _____, comprising a standardized set of terms/rules used in E-Commerce/Trade.

- (a) NASSCOM
(b) World Trade Organization (WTO)
(c) International Chamber of Commerce (ICC)

- (d) International Monetary Fund (IMF)
(e) World Economic Forum (WEF)

153. What does the "R" in LIDAR stand for?

- (a) Radar (b) Reflection (c) Recognition
(d) Remote (e) Ranging

154. What is the minimum capital requirement for setting up a Small Finance Bank (SFB) in India?

- (a) ₹50 crore (b) ₹100 crore
(c) ₹200 crore (d) ₹500 crore
(e) ₹1000 crore

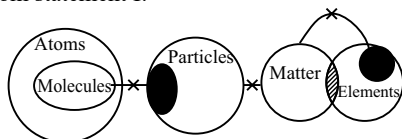
155. Which company has secured orders from two regional banks — Kerala Gramin Bank and Karnataka Gramin Bank — for its banking Superapp and digital transformation?

- (a) Network People Services Technologies Limited (NPST)
(b) Tata Consultancy Services (TCS)
(c) Infosys Limited
(d) Wipro Limited
(e) HCL Technologies

Answer Key & Solutions

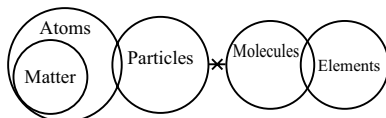
1. (e) According to the question,

From statement I:



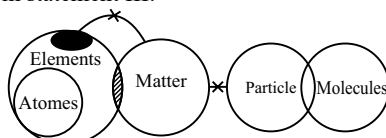
There is no direct relation between matter and molecules.

From statement II:



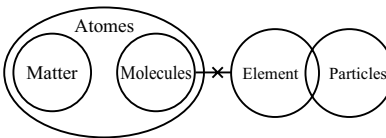
Some atoms are definitely matter

From statement III:



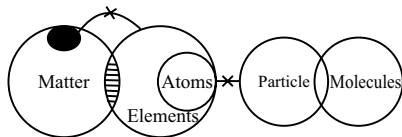
Some matter being molecules is a possibility.

From statement IV:



Some atoms are definitely matter.

From statement V:



There is no direct relation between matter and molecules.

Therefore, none of the statements satisfy the given conclusions..

Solutions (2-6):

Numbers: Even numbers are arranged in decreasing order from the left end and odd numbers are arranged in decreasing order from the right end simultaneously.

Words: Words are arranged in alphabetical order by adding previous letter of first alphabet to the left side of the word and by adding previous letter of the last alphabet to the right side of the word.

Input: ETU 75 GTQ 14 RTV 22 NTS 32 MTW 67 FSN 19 CER

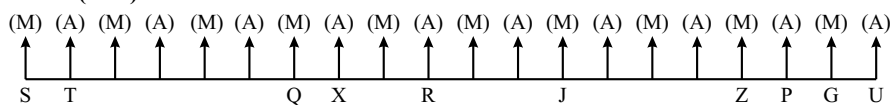
Step 1: 32 ETU GTQ 14 RTV 22 NTS MTW 67 FSN 19 CER 75

Step 2: 22 32 ETU GTQ 14 RTV NTS MTW FSN 19 CER 75 67

Step 3: 14 22 32 ETU GTQ RTV NTS MTW FSN CER 75 67 19

Step 4: 14 22 32 BCERQ ETU GTQ RTV NTS MTW FSN 75 67 19

Solutions (7-10):



7. (a) 4 persons are seated between P, the third Associate to the right of J and J, the fourth Manager from the extreme right end of the row.

8. (c) J is seated exactly in the middle of R and the person, who sits immediately to the left of Z.

9. (d) The number of Managers sitting between X and P is more than two is definitely true.

Step 5: 14 22 32 BCERQ DETUT GTQ RTV NTS MTW FSN 75 67 19

Step 6: 14 22 32 BCERQ DETUT EFSNM GTQ RTV NTS MTW 75 67 19

Step 7: 14 22 32 BCERQ DETUT EFSNM FGTQP RTV NTS MTW 75 67 19

Step 8: 14 22 32 BCERQ DETUT EFSNM FGTQP LMTWV RTV NTS 75 67 19

Step 9: 14 22 32 BCERQ DETUT EFSNM FGTQP LMTWV MNTSR RTV 75 67 19

Step 10: 14 22 32 BCERQ DETUT EFSNM FGTQP LMTWV MNTSR QRTVU 75 67 19

2. (a) Clearly, there is no element between 'RTV' and 'NTS' in the step 8.

3. (d) Clearly, 6 elements are there between BCERQ and 75 in the step 9.

4. (d) Clearly, 'EFSNM' is sixth from the left in the step 7.

5. (a) Clearly, 'BCERQ' is second to the left of 'GTQ' in the step 4.

6. (e) Clearly, step 10 is the last step.

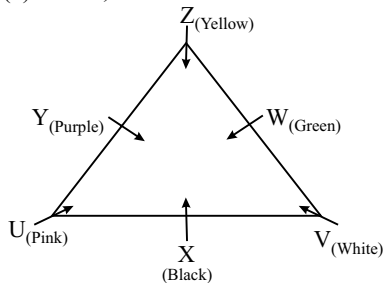
10. (b) Immediate left is the position of the one who sits third to the left of the 5th Manager from left end of the row with respect to Q.

11. (d) From option (a), L would be brother of C, so L would be male, so option (a) can be eliminated.

In option (b) and (c), L is sister of X, so both can be eliminated.

In option (d),
 $L(-)=G(+)-X(+)=C(-)$
 So, L is sister-in-law of X.

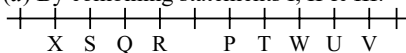
12. (a) From I, II and III:



W likes Green colour.

Hence, data in statements I, II and III together are sufficient to answer the question.

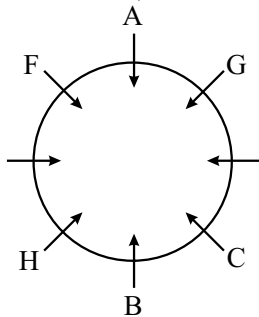
13. (a) By combining statements I, II & III:



So, three persons sit between P and V.

Hence, statements I, II and III together are sufficient to answer the question.

14. (b) From statements I, II and III:



But, position of D and E can't fix as C is not an immediate neighbour of D and F sits between E and A.

So, on combining the given statements we cannot find the answer.

15. (b) From I, II and III:

Words	Code
Ninja	cd
Turtle	mn
Game	dk
Famous	ef
In	pq
The/world	lm/dt
Now	Ex

As, exact word for code 'dt' can't find.

Hence, data in statements I, II and III are not sufficient to answer the question.

16. (c) Word: COLLABORATION

Arrangement: B P K K B A P Q B S J P M

Word: DETERMINATION

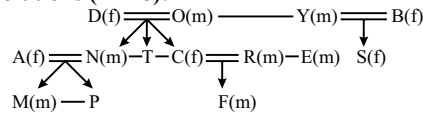
Arrangement: E F S F Q N J N B S J N N

Combined: A (B) B B (B) E F F J J K K M N N N N P P P Q Q S S S

Left → 2nd → 3rd Right

Hence, B is 3rd to the right of the 2nd letter from the left end.

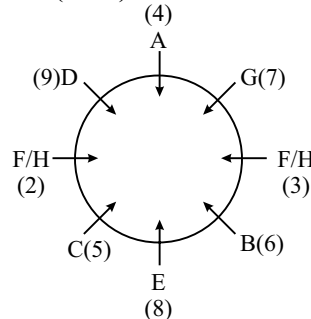
Solutions (17-18):



17. (c) S is the niece of O.

18. (a) N is the brother-in-law of R.

Solutions (19-22):



19. (d) G and the person holding card numbered 3 are seated adjacent to each other is the correct statement.

20. (e) Either F or H is seated immediately to the left of the person holding the card numbered 7, so the answer is can't be determined.

21. (d) The card number held by the person seated opposite to B is 9.

22. (d) If the sum of the card numbers held by A and H is 7, then the difference between the card numbers held by C and the one who is sitting second to the left of E is $= 5 - 2 = 3$.

As card numbered of H is $= 7 - 4 = 3$.

Solutions (23-24):

The given arrangement,

& F A # L K 4 \$ 9 G P @ 8 B S M @ € 5 D % L β K 4 A 9 @ S W ¥ 6 B & 8 # J

After applying step I, we get:

& # % β @ # F A L K 4 \$ 9 G P @ 8 B S M @ € 5 D L K 4 A 9 S W ¥ 6 B & 8 J

After applying step II, we get:

& # % β @ # F A L K 4 \$ 9 G @ 8 B S @ € 5 D L K 4 A 9 S ¥ 6 & 8 J B M P W

After applying step III, we get:

& # % β @ # F A L K 4 9 9 8 6 5 4 \$ G @ B S @ € D L K A S ¥ & 8 J B M P W

After applying step IV, we get:

4 9 9 8 6 5 4 \$ G @ B S @ € D L K A S ¥ & 8 J B M P W & # % β @ # F A L K

23. (e) Sixteen elements are between two A's after applying step IV.

24. (d) The 21 element from the left end is '&', 7 element to the left of '&' is '€'.

Solutions (25-26):

25. (a) Statement I: Follows (The passage explicitly states that Microsoft and OpenAI have formed a strategic alliance

to advance the frontier of AI research and development, supporting the conclusion that they have joined forces for this purpose.)

Statement II: Follows (The passage mentions that the partnership leverages Microsoft's vast resources and technological prowess, alongside OpenAI's cutting-edge AI research, supporting the conclusion that their collaboration combine these respective strengths.)

Statement III: Doesn't follow (The passage does not specify the focus of the partnership on consumer-oriented AI products and services. Instead, it mentions collaborations aimed at tackling complex challenges across various domains, suggesting a broader scope beyond consumer-oriented applications.)

Hence, Only I and II follow.

26. (a) Statement I: Follows (The passage mentions collaborations on projects aimed at tackling complex challenges across various domains, suggesting that expanding the scope of collaborative projects to address global challenges aligns with their partnership objectives.)

Statement II: Doesn't follow (The passage does not indicate that the partnership's primary objective is commercialization for profit-driven ventures. Instead, it highlights collaborations on projects aimed at pushing the boundaries of AI capabilities and addressing societal challenges.)

Statement III: Doesn't follow (There is no indication in the passage that collaborative efforts between Microsoft and OpenAI should be discontinued due to conflicting research priorities or organizational objectives. Instead, it emphasizes their partnership in advancing AI research and development.)

Hence, only I follows.

27. (b) However, when trying to create a valid word using all five letters D, U, T, M, S, there is no common meaningful English word that uses all five letters exactly once.

Solutions (28-32):

Dates (March)	Persons
20	D
21	A
22	Vacant
23	C
24	Vacant
25	B
26	Vacant
27	E
28	F

28. (e) 24th and 26th are vacant dates.

29. (b) D was born on 20th March.

30. (d) No one was born between B and C.

31. (c) D was born before C is true statement.

32. (d) No one was born on 22nd March.

33. (d) **From statement I:** Doesn't follow (Imposing heavy tariffs on semiconductor imports contradicts the statement because it would disrupt the global electronics industry's supply chain instead of acknowledging the critical role semiconductor facilities play in it.)

From statement II: Doesn't follow (Neglecting investments in research and development and relying solely on outdated semiconductor technologies also contradicts the statement. Semiconductor facilities need to continually innovate to keep pace with technological advancements and meet the demands of the electronics industry.)

Hence, neither I nor II follows.

34. (b) **Argument I:** While lifting the ban may promote financial inclusion, it could also expose users to potential security threats and risks.

This is not a strong argument because it's contradictory.

Argument II: Maintaining the ban protects users from potential fraudulent activities and data breaches, ensuring their financial security.

This is a strong argument. It provides a clear and relevant reason focused on user safety and fraud prevention.

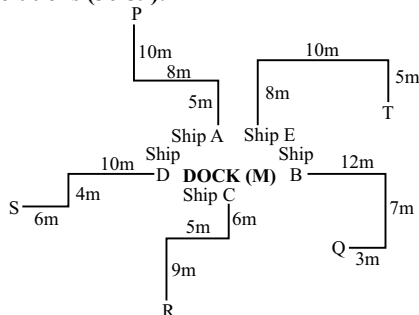
Hence, only Argument II is strong.

35. (e) **Assumptions I:** Implicit (This assumption, though not explicitly stated, is implied within the statement. It suggests that immigration policies need to include measures to regulate the flow of migrants into a country, balancing the need for security with maintaining legal immigration channels to facilitate orderly migration.)

Assumptions II: Implicit (Similarly, this assumption is not explicitly stated but is implied within the statement. It suggests that immigration policies that embrace diversity and inclusivity have positive societal impacts. Such policies can help fill labour shortages, promote economic growth, and foster innovation by welcoming immigrants and their diverse skills and perspectives.)

Hence, both I and II are implicit.

Solutions (36-39):



36. (b) S is in north-west direction of R.

37. (d) The total distance covered by Q and R in their journey is

$$= (12 + 7 + 3) + (6 + 5 + 9) = 42\text{m}.$$

38. (c) T is in north-east of Q is true statement.

39. (d) The shortest distance between the initial and final position of ship B is

$$= \sqrt{9^2 + 7^2} = \sqrt{81 + 49} = \sqrt{130}\text{ m}$$

Solutions (40-41):

F	B	E
I		G

Car 2

D	X	H
A		C

Car 1

40. (d) No one is seated in 1d.

41. (c) Except C, all of them are seated in Car 2.

Solutions (42-45):

Members	B H L	E O M	D J R	A K I	E N Q	G C P
Groups	Red	Brown	Orange	Olive	Purple	Violet

42. (e) Twelve students are sitting to the right of group member O.

43. (b) B and H belong to the Red group.

44. (c) J and N are in Orange group is odd from the rest, because J and N are from different groups.

45. (e) Brown, Purple, Olive and Orange groups are between Red group and Violet group.

46. (b) The passage mentions that Japan's low fertility rate and high life expectancy are key factors contributing to its ageing population. Immigration and technological innovation are not emphasized as primary causes.

47. (a) The passage highlights that shifting to community-based care addresses the strain on healthcare resources and allows elderly individuals to receive care in familiar settings.

48. (b) The passage discusses how Japan's ageing population has led to a reduced working-age population, and the government encourages older individuals to stay employed to mitigate the impact.

49. (c) The passage mentions Japan's historically conservative stance on immigration and highlights the Specified Skilled Worker visa program as a way to address labor shortages.

50. (c) The passage discusses how Japan uses robotic assistants with AI to alleviate caregiver burdens and help the elderly maintain independence.

51. (c) The passage explains that pension reforms include raising eligibility ages and promoting private retirement savings to ensure the system's sustainability.

52. (a) The passage mentions that intergenerational programs help reduce isolation in the elderly while fostering mutual support between age groups.

53. (b) The passage describes these centers as initiatives that offer part-time work or volunteering opportunities to older individuals.

54. (b) The passage states that Japan's public debt is one of the highest globally, posing a significant challenge despite its efforts to manage the ageing population.

55. (b) The passage explains that betting apps use gamified elements, such as streaks and rewards, which mimic video games, making them more appealing to younger audiences.

56. (c) The passage states that these advertisements frame gambling as harmless or glamorous, often ignoring the risks involved.

57. (b) The passage mentions that these ads manipulate cognitive biases, such as making users believe they can influence outcomes, making the apps more appealing.

58. (c) The passage discusses how gambling habits can create conflicts within families and increase peer pressure to participate in such activities.

59. (c) The passage highlights that youth often get trapped in a cycle of trying to recover losses, exacerbating their financial vulnerability.

60. (b) The passage emphasizes the need for stricter regulations, alongside awareness and education, to mitigate the harmful effects of such advertisements.

61. (b) The passage mentions that while manufacturing has historically driven China's growth, recent trends indicate a shift toward services and domestic consumption, supported by e-commerce and technology.

62. (b) The passage explains that China faces issues such as environmental concerns and other growth challenges. The government is focusing on green energy and innovation as solutions to sustain economic growth.

63. (d) The passage states that future COPs must focus almost exclusively on ensuring delivery and holding laggards accountable. The latest round of discussions (COP27 in Egypt) began with little hope and ended (two days beyond schedule) with a faint glimmer. As per the context of the passage both the words 'discussions' and 'dialogue' are suitable options to fill the blank (A). Hence the correct answer choice is option (d).

64. (c) The passage states that future COPs must focus almost exclusively on ensuring delivery and holding laggards accountable. The 197 Parties to the UN Framework Convention on Climate Change (UNFCCC) decided to make it possible to finance loss and damage for those who most need it. However, the objective of limiting temperature rise to levels that could minimize loss. Therefore, as per the context of the passage 'Minimize' is the correct word to fill the blank (B). Hence the correct answer choice is option (c).

Exaggerate: To make something worse, especially disease or problem.

Exacerbate: To make something worse especially disease or problem

65. (a) The passage states that future COPs must focus almost exclusively on ensuring delivery and holding laggards accountable. Therefore, as per the context of the passage 'Vulnerable' is the correct word to fill the blank (C). Loss and damage (L&D) is the term used to describe the negative effects that a changing climate has on vulnerable communities and nations. These effects include an increase in the frequency and severity of natural disasters and extreme weather events, as well as slow-onset temperature rise, sea-level rise, and desertification. 'Distinctive' and 'Rapport' are not suitable words to fill the blank as these are out of the context of the passage. Hence the correct answer choice is option (a).

Distinctive: Clearly different from others and therefore easy to recognise

Rapport: A friendly relationship in which people understand each other very well.

66. (e) The passage states that future COPs must focus almost exclusively on ensuring delivery and holding laggards accountable. Rich nations had opposed L&D payments for years out of concern that the previous emissions might result in significant compensation demands. They were under so much pressure that they had to accept their accountability. Impediment means Something that makes it difficult for a person to progress or something that makes speaking difficult. 'Impediment' therefore is not the correct word to fill the blank (D). Onus means Responsibility. Therefore, as per the context of the passage 'Onus' and 'Responsibility' both words are suitable words to fill the blank (D). Hence the correct answer choice is option (e).

67. (b) The passage states that future COPs must focus almost exclusively on ensuring delivery and holding laggards accountable. This committee has been entrusted with setting up institutional arrangements, locating and expanding sources of funding, and coordinating with current funding arrangements by COP28

in the UAE the following year in order to operationalize the financial arrangements. Therefore, as per the context of the passage 'Arrangements' is the correct word to fill the blank (E). 'Advancement' and 'Channelization' are not the suitable words to fill the blank as these words are out of context of the passage. Hence the correct answer choice is option (b).

68. (b) Let's examine the significance of each alternative one by one using the elimination procedure.

Possibility 1: Sharan and Ram have incredible voices, similarly they will go far in their music career.

Clearly, the above formulation is not grammatically valid, it does not form a meaningful statement.

Possibility 2: Sharan and Ram have incredible voices, otherwise they will go far in their music career.

The preceding formulation is not grammatically valid, it does not form a meaningful statement.

Possibility 3: Sharan and Ram have incredible voices, undoubtedly they will go far in their music career.

The above synthesis is clearly grammatically and contextually accurate, and it keeps the flow going.

69. (e) In legal language, someone is always punished under a certain section of law. Only (iii) acts as the right connector and hence, option (e) is the right answer.

70. (b) The mentioned person changed his name after an incident.

So, out of the given connectors only the one given in option (b) will be correct in the context.

Therefore, the best option is (b).

71. (b) Both 'however' and 'meanwhile' will be the best to join the two separate sentences given above. Though will not be correct grammatically and hence, that can be eliminated.

Therefore, option (b) is the correct answer.

72. (d) The blank seems to be filled by the subject/topic/theme that twitter does not want to move to other platforms which in turn will increase its advertising revenue. Part (B) and (D) provide hints for the theme of the paragraph. Among the options, option (d) complements the given information in the paragraph. Hence, option (d) is the correct answer.

Option (a) does not contribute contextually to the passage.

Option (b) is incorrect as it talks about twitter's paid subscriber count.

Option (c) does not contribute to the passage as it talks about the twitter's dominance in the market.

73. (c) (a) Option (a) lists the specific financial services but does not relate with the context of the sentence.

(b) Option (b) suggests that financial inclusion targets previously inaccessible financial systems, but this is not the primary focus of the given statement.

(c) Option (c) accurately summarizes the main goal of financial inclusion, which is to ensure individuals and communities have access to a broad range of financial services and resources.

(d) Option (d) mentions the consequences of lacking financial inclusion, which can limit investments in education, healthcare, and entrepreneurship, but this is not the central theme of the statement.

Hence the correct option choice is (c).

74. (b) Option (b) is the correct choice - "grapples with its far-reaching implications."

The phrase "grapples with its far-reaching implications" means that the world is struggling with the significant and wide-ranging consequences of climate change. It emphasizes that climate change is not just a localized or isolated problem but a complex issue with profound effects on various aspects of the environment, society, and the economy worldwide.

The other options, while relevant to the issue of climate change, do not accurately capture the overall context of the sentence.

Option (a) addresses specific aspects of climate change but does not convey the comprehensive and challenging nature of the issue. Option (c) highlights some areas of focus, but it is not as encompassing as the phrase in option (b). Option (d) is closely related to the consequences of climate change, but it lacks the broader perspective captured by the term "far-reaching implications."

Hence, the correct option choice is (b).

75. (d) Option (a) focuses on generating returns from investments but doesn't cover the broader impacts on economic development, innovation, and projects.

Option (b) highlights promising growth potential in entrepreneurial endeavors, but it omits the wider context of economic development and various projects.

Option (c) mentions facilitating diverse initiatives and ventures, yet it doesn't explicitly encompass the effects on economic development and technological innovation.

Option (d) accurately captures the statement's main points about investments impacting economic development, technological innovation, and project realization.

76. (b) The passage states: "New energy-efficient technologies, most likely harnessing artificial intelligence, will dominate nearly all human activities from farming to heavy industry." It shows that new energy will be dominating human activities. It does not, however, state that

AI will be the only way of harnessing new energy. As alternatives, the passage suggests "nuclear fusion and other new energy sources". Thus, statement A cannot be inferred from the passage. The quoted sentence also shows how statement B can be correctly inferred from the passage.

The passage does not state nuclear fusion as a method of harnessing clean energy. And it also does not link AI and nuclear fusion as competitive methods of harnessing energy. Thus, statement C is an assumption, not an inference.

77. (e) Here, D is the introductory part of the sentence that states the experiences faced by the world. D is followed by E that states the benefits of preserving ecosystems. A follows E. further A is followed by B which states regarding natural hazards. C follows B and concludes the sentence. The correct arrangement will be DEABC. Therefore, option (e) is the correct answer. The rearranged sentence will be:

There are many experiences around the world which point to the potential benefits of protecting and preserving ecosystems such as wetlands, forests, lakes, and coastal areas in reducing physical exposure to natural hazards such as floods, landslides or avalanches by serving as buffers.

78. (b) Here, A is the introductory part of the sentence that states the situation of people during lockdown. A is followed by B. B follows C. further C is followed by E which states the places where one feels less stressed. D follows E and concludes the sentence. The correct arrangement will be ABCED. Therefore, option (b) is the correct answer.

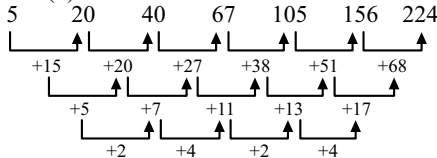
The rearranged sentence will be:

Even during the lockdown, people cooped up in homes with less than usual activity were advised by psychologists to take time off and relax in places close to home where one feels less stressed, as it is important for mental and physical wellbeing.

79. (b) The positions of words in the given paragraph are incorrect. Hence, swapping the words in their correct positions will frame a grammatically correct and contextually meaningful sentence. The word (1) will be swapped with the word (2) and (3) with the (5). Hence, the correct option choice is (b).

80. (b) The word adventure refers to an exciting or daring experience, often involving exploration or a journey. Among the given options, exploration best matches the meaning of adventure. Other options like "danger" or "routine" do not convey the same sense of excitement or thrill.

81. (d) From series I:



Value of P = 140

From series II: 13, 20, 30, 98, 119, 334

$$20 - 13 = 7 = 2^3 - 1$$

$$30 - 20 = 10 = 3^2 + 1$$

$$93 - 30 = 63 = 4^3 - 1$$

$$119 - 93 = 26 = 5^2 + 1$$

$$334 - 119 = 215 = 6^3 - 1$$

Value of Q = 98

Difference between P and Q = 42

82. (d) We know that, Dividend = Divisor $\times$ Quotient + Remainder

So,

$$6x^2 + 29x + P = (3x + 1) \times (2x + 9) + 19$$

$$\Rightarrow 6x^2 + 29x + P = 6x^2 + 27x + 2x + 9 + 19$$

$$\Rightarrow P = 28$$

83. (a) $6x^2 + 29x + 28 = 0$

$$\Rightarrow 6x^2 + 21x + 8x + 28 = 0$$

$$\Rightarrow 3x(2x + 7) + 4(2x + 7) = 0$$

$$\Rightarrow (2x + 7)(3x + 4) = 0$$

$$\Rightarrow x = -\frac{7}{2}, -\frac{4}{3}$$

Hence, largest root is $-\frac{4}{3}$, i.e., $-1\frac{1}{3}$.

84. (a) Let, marked price per article be M

For 160 articles:

Selling price per article = 0.8M

$$\text{Total selling price} = 160 \times 0.8M = 128M$$

For 80 articles:

Selling price per article

$$= 0.8M \times 0.75 = 0.6M$$

$$\text{Total} = 80 \times 0.6M = 48M$$

$$\text{Total revenue (Z)} = 128M + 48M = 176M$$

Given 10% profit,

$$176M = 1.1 \times \text{Cost price}$$

$$\Rightarrow \text{Cost Price} = \frac{176M}{1.1} = 160M$$

Total selling price without discount

$$= 240M$$

$$\text{Profit} = 240M - 160M = 80M$$

$$\therefore \text{Profit \%} = \left(\frac{80M}{160M} \right) \times 100 = 50\%$$

85. (b) Container X:

Quantity of milk

$$= 140 \times \left(1 - \frac{250}{700} \right) = 90 \text{ liters}$$

Quantity of water = 30 liters

Ratio of milk to water = 90 : 30 = 3 : 1

40% of the mixture is taken out

Quantity of milk

$$= 90 - (90 + 30) \times \frac{40}{100} \times \frac{3}{4} = 54 \text{ liters}$$

Quantity of water

$$= 30 - (90 + 30) \times \frac{40}{100} \times \frac{1}{4} = 18 \text{ liters}$$

Container Y:

Quantity of soda

$$= 80 \times \frac{100 - 62.5}{100} = 30 \text{ liters}$$

Required percentage

$$= \frac{18}{30} \times 100 = 60\%$$

Solutions (86-90):

Number of employees, in account department
 = 20% of 6000 = 1200

Number of employees who got promotion in account department = 30% of 1200 = 360

Similarly, we can calculate number of employees and number of employees, who got promotion for remaining departments.

Department	Total number of employees	Number of employees got promotion
Account	1200	360
HR	1500	180
IT	1800	240
Production	900	180
Sells	600	240

86. (d) Total number of male employees, who promoted in the HR department
 = 180 - 120 = 60

Total number of male employees in the HR department = $\frac{7}{12} \times 1500 = 875$

$$\text{Required ratio} = 875 : 60 = 175 : 12$$

87. (a) The number of female employees in the production department

$$= \frac{3}{5} \times 900 = 540$$

The number of female employees who get promotions in the production department

$$= \frac{5}{9} \times 180 = 100$$

$$\text{Required \%} = \frac{100}{540} \times 100 = 18.52\%$$

88. (d) The percentage of the number of employees, who get promotion from the

$$\text{HR department} = \frac{180}{1500} \times 100 = 12\%$$

The percentage of the number of employees, who get promotion from the

$$\text{IT department} = \frac{240}{1800} \times 100 = 13.33\%$$

$$\text{Required difference} = 13.33\% - 12\% = 1.33\%$$

89. (b) Required % = $\frac{360 + 240 + 240}{1200 + 1800 + 600}$

$$= \frac{840}{3600} \times 100 = 23.33\%$$

90. (c) Required difference

$$= \frac{1800 + 900 + 600}{3} - \frac{240 + 180 + 240}{3}$$

$$= 1100 - 220 = 880$$

91. (b) Let total number of balls in the bag
 $= 24x$
 Probability of drawing a blue colour ball
 $= \frac{5}{24}$

Total blue colour balls $= 5x$

Total red and green colour balls

$$= 24x - 5x = 19x$$

$$\text{Total red colour balls} = 19x \times \left(\frac{9}{19}\right) = 9x$$

Total green colour balls

$$= 19x \times \left(\frac{10}{19}\right) = 10x$$

From I: Average of total number of balls of all the three colours-

$$\frac{5x + 9x + 10x}{3} = 8$$

$$\Rightarrow 24x = 24 \Rightarrow x = 1$$

Total balls in the bag $= 24x = 24$

Statement I alone is sufficient.

From II: Probability that two drawn balls are of green colour $= \frac{{}^{10x}C_2}{{}^{24x}C_2} = \frac{15}{92}$

$$\Rightarrow \frac{5x(10x-1)}{12x(24x-1)} = \frac{15}{92}$$

$$\Rightarrow 23 \times (10x - 1) = 9(24x - 1)$$

$$\Rightarrow 230x - 23 = 216x - 9$$

$$\Rightarrow 14x = 14 \Rightarrow x = 1$$

Total balls in the bag $= 24x = 24$

Statement II alone is sufficient.

From III: Probability that one ball is drawn and that ball is either red or blue-

$$\frac{{}^{9x}C_1 + {}^{5x}C_1}{{}^{24x}C_1} = \frac{7}{12}$$

$$\Rightarrow \frac{9x + 5x}{24x} = \frac{7}{12} \Rightarrow \frac{14}{24} = \frac{7}{12} \Rightarrow 1 = 1$$

Statement III alone is not sufficient.

92. (d) **From I:**

$$\text{Syrup} = \frac{a}{a+b} \times 54$$

$$\text{Water} = \frac{b}{a+b} \times 54$$

$$\left[\frac{a}{a+b} \times 54 + 10 \right] \div \left[\frac{b}{a+b} \times 54 + 6 \right] = \frac{4}{3}$$

$$\Rightarrow 3 \times \left[\frac{a}{a+b} \times 54 + 10 \right] = 4 \times \left[\frac{b}{a+b} \times 54 + 6 \right]$$

$$\Rightarrow \frac{162a}{a+b} + 30 = \frac{216b}{a+b} + 24 \Rightarrow \frac{a}{b} = 5 : 4$$

$$\Rightarrow \frac{162a}{a+b} + 30 = \frac{216b}{a+b} + 24 \Rightarrow \frac{a}{b} = 5 : 4$$

From II:

$$\text{Amount of milk} = \frac{7}{7+5} \times 60$$

$$= \frac{7}{12} \times 60 = 35 \text{ litres}$$

$$\text{Amount of water} = \frac{5}{7+5} \times 60$$

$$= \frac{5}{12} \times 60 = 25 \text{ litres}$$

$$\frac{35 - \frac{7z}{12}}{25 - \frac{5x}{12} + z} = \frac{7}{23}$$

$$\Rightarrow 23 \times \left(35 - \frac{7z}{12} \right) = 7 \times \left(25 - \frac{5z}{12} + z \right)$$

$$\Rightarrow 23 \times \frac{420 - 7z}{12} = 7 \times \frac{300 - 5z + 12z}{12}$$

$$\Rightarrow 23 \times (420 - 7z) = 7 \times (300 + 7z)$$

$$\Rightarrow 9660 - 161z = 2100 + 49z$$

$$\Rightarrow 161z + 49z = 9660 - 2100$$

$$\Rightarrow 210z = 7560$$

$$\Rightarrow z = \frac{7560}{210} = 36$$

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$$\text{So, height} = \frac{11340}{20 \times 21} = 27$$

$$\text{I. } 3m = 3 \times 9 = 27$$

$$\text{II. } \frac{n}{4} = \frac{32}{4} = 8 \neq 27$$

$$\text{III. } n - m - 15 = 32 - 9 - 15 = 8 \neq 27$$

Solutions (95-98):

First we need to calculate the value of m and n .

$$K^2 - 32P + 252 = 0$$

$$\Rightarrow (K - 18)(K - 14) = 0$$

$$\Rightarrow K = 18, 14$$

So, value of $n = 4 \times 18 = 72$

For calculating value of m ,

$$140 - 2^2 = 136$$

$$136 + 5^2 = 161$$

$$161 - 11^2 = 40$$

$$40 + 17^2 = 329$$

$$329 - 23^2 = -200$$

So, value of $m = 2 \times 40 = 80$

$$\text{So, cost price of T} = \left(4a + \frac{80}{2} \right) = (4a + 40)$$

$$\text{Marked price of T} = \left(13a - \frac{72}{2} + 6 \right) = (13a - 30)$$

Now,

$$(13a - 30) - (4a + 40) = 1280$$

$$\Rightarrow 9a = 1350$$

$$\text{So, value of } a = \frac{1350}{9} = 150$$

$$\text{Profit earned on article S} = \frac{1280}{128} \times 100 = ₹1000$$

Selling price of article S,

$$= (5 \times 80 + 7 \times 72 + 4 \times 150 + 296) = ₹1800$$

So, cost price of article S $= 1800 - 1000 = ₹800$

$$\text{So, } (5b - m - n - 48) = 800$$

$$\Rightarrow 5b = 800 + 48 + 72 + 80$$

$$\Rightarrow \text{Value of } b = \frac{1000}{5} = 200$$

Now, we have the following known values

Article	Cost price	% Profit	Marked price
P	—	40%	2400
Q	—	—	1728
R	—	20%	—
S	800	—	—
T	640	—	1920

95. (c) Cost price of P $= 187.5\%$ of 640
 $= ₹1200$

Selling price of P $= 1200 \times 1.4 = ₹1680$

$$\text{Discount} = \frac{2400 - 1680}{2400} \times 100 = 30\%$$

$$\text{So, } (2L - 210) = 30$$

$$\text{Value of } L = \frac{240}{2} = 120$$

$$\text{I. } 1.5m = 1.5 \times 80 = 120$$

$$\text{II. } (b - n - 8) = 200 - 72 - 8 = 120$$

$$\text{III. } 2.4(b - a) = 2.4 \times (200 - 150) = 120$$

IV. 240

Only I, II, and III follows.

96. (b) Cost price of R = $80 + 1920 = ₹2000$
 MP of R = $2000 \times 1.2 + 400 = ₹2800$
 So, discount % = $\frac{400}{2800} \times 100 = \frac{100}{7}\%$
 Now, $\frac{100}{7} = \frac{5L}{14} \Rightarrow L = 40$

97. (a) Discount % given on article Q = $\frac{50}{3}\%$
 So, profit % on article Q
 $= 3 \times \frac{50}{3} = 50\%$
 Selling price of article Q
 $= \frac{5}{6} \times 1728 = ₹1440$
 Cost price of article Q = $\frac{1440}{1.5} = ₹960$
 Cost price of article S = ₹800
 Required difference = $960 - 800 = ₹160$

98. (d) Discount % on T = $(200 - 150)\% = 50\%$
 So, selling price of T = $1920 \times 50\% = ₹960$
 Profit% on T
 $(L\%) = \frac{960 - 640}{640} \times 100 = 50\%$
 MP of S = $2400 - 400 = ₹2000$
 Selling price of S = ₹1800
 Discount % on S
 $= \frac{2000 - 1800}{2000} \times 100 = 10\%$
 Required value = $(50\% - 10\%) = 40\%$

Solutions (99-101):

Number of caramel candies in pouch D is 2 more than that in pouch A.
 So, number of caramel candies in pouch A = $10 - 2 = 8$

	Orange candy	Pineapple candy	Caramel candy	coffee candy
Pouch A	12	a	8	b
Pouch B	c	24	d	10
Pouch C	e	f	16	g
Pouch D	h	f	10	n
Pouch E	p	10	p	q
Total			69	69

For pouch A:

When one candy is picked at random from pouch A, then the probability of getting one coffee candy is $\frac{1}{3}$

$$\Rightarrow \frac{b}{(20 + a + b)} = \frac{1}{3} \Rightarrow 3b = 20 + a + b$$

$$\Rightarrow 2b - a = 20 \quad \dots (i)$$

And, when two candies are picked at random from pouch A, then the probability of getting both pineapple candies are $\frac{1}{22}$.

$$\Rightarrow \frac{[a(a-1)]}{[(a+b+20)(a+b+19)]} = \frac{1}{22}$$

$$\Rightarrow \frac{[a(a-1)]}{[3b(3b-1)]} = \frac{1}{22}$$

$$\Rightarrow 22a(a-1) = 3b(3b-1)$$

$$\Rightarrow 22 \times (2b-20) \times (2b-21) = 3b(3b-1)$$

$$\Rightarrow b = 15$$

So, $a = 2 \times 15 = 20 = 10$
 So, total number of candies in pouch A = $12 + 10 + 8 + 15 = 45$
 For pouch E:
 When one candy is picked at random from pouch E, then the probability of getting one orange candy is $\frac{1}{6}$.

$$\Rightarrow \frac{p}{(2p+q+10)} = \frac{1}{6} \Rightarrow 6p = 2p + q + 10$$

$$\Rightarrow 4p - q = 10 \quad \dots (ii)$$

When one candy is picked at random from pouch E, then the probability of getting one pineapple candy is $\frac{1}{3}$.

$$\Rightarrow \frac{10}{2p+q+10} = \frac{1}{3}$$

$$\Rightarrow 2p + q + 10 = 30 \Rightarrow 2p + q = 20 \quad \dots (iii)$$

Solving equation (ii) and (iii), we get
 $p = 5$ and $q = 10$
 For pouch B:
 So, $d = 69 - 8 - 16 - 10 - 5 = 30$
 When one candy is picked at random from pouch B, then the probability of getting either orange candy or caramel candy is $\frac{25}{42}$.

$$\Rightarrow \frac{(c+d)}{(c+d+24+10)} = \frac{25}{42} \Rightarrow \frac{(c+30)}{(c+64)} = \frac{25}{42}$$

$$\Rightarrow c = 20$$

So, total number of candies in pouch B = $20 + 24 + 30 + 10 = 84$

For pouch C:

When one candy is picked at random from pouch C, then the probability of getting neither caramel nor coffee is $\frac{1}{2}$.

$$\Rightarrow \frac{(e+f)}{(e+f+g+16)} = \frac{1}{2}$$

$$\Rightarrow 2(e+f) = e+f+g+16$$

$$\Rightarrow e+f-g = 16 \quad \dots (iv)$$

When one candy is picked at random from pouch C, then the probability that the candy picked is not caramel flavor is $\frac{4}{5}$.

$$\Rightarrow \frac{(e+f+g)}{(e+f+g+16)} = \frac{4}{5}$$

$$\Rightarrow 5(e+f+g) = 4(e+f+g) + 64$$

$$\Rightarrow e+f+g = 64 \quad \dots (v)$$

So, total = $64 + 16 = 80$
 So, $e+f = 40 \quad \dots (vi)$
 And $g = 24$

For pouch D:

So, $n = 69 - 15 - 10 - 24 - 10 = 10$

When one candy is picked at random from

pouch D, then probability of getting coffee candy is $\frac{1}{5}$.

$$\Rightarrow \frac{10}{\text{total}} = \frac{1}{5} \Rightarrow \text{total} = 50$$

So, when two candies are picked at random from pouch D, then the probability that the none of the candies is orange flavor is $\frac{17}{35}$.

$$\Rightarrow \frac{[(f+20)(f+19)]}{[(h+f+20)(h+f+19)]} = \frac{17}{35}$$

$$\Rightarrow (f+20)(f+19) = \frac{17}{35} \times 50 \times 49 = 1190$$

$$\Rightarrow f = 15 \text{ or } -54$$

So, from equation (vi), we get
 $e = 40 - 15 = 25$

Tabulating the above data:

	Orange	Pineapple	Caramel	Coffee	Total
A	12	10	8	15	45
B	20	24	30	10	84
C	25	15	16	24	80
D	15	15	10	10	50
E	5	10	5	10	30
Total	77	74	69	69	289

99. (d) If 5 orange candies from pouch A are shifted to pouch C and 2 caramel candies from pouch D are shifted to pouch A.

Pouch A:

Orange candies = $12 - 5 = 7$

Pineapple candies = 10

Caramel candies = $8 + 2 = 10$

Coffee candies = 15

Total = $7 + 10 + 10 + 15 = 42$

Pouch C:

Orange candies = $25 + 5 = 30$

Pineapple candies = 15

Caramel candies = 16

Coffee candies = 24

Total = $30 + 15 + 16 + 24 = 85$

Pouch D:

Orange candies = 15

Pineapple candies = 15

Caramel candies = $10 - 2 = 8$

Coffee candies = 10

Total = $15 + 15 + 8 + 10 = 48$

Quantity I:

$$\text{Required probability} = \frac{{}^8C_2}{{}^{48}C_2} = \frac{8 \times 7}{48 \times 47}$$

$$= \frac{7}{282}$$

Quantity II:

Required probability

$$= \frac{({}^{10}C_2 \times {}^7C_1 + {}^{10}C_2 \times {}^{10}C_1 + {}^{10}C_2 \times {}^{15}C_2)}{{}^{42}C_3}$$

$$= \frac{\left[\left(\frac{10 \times 9}{2} \right) \times (7 + 10 + 15) \right]}{\left[\frac{(42 \times 41 \times 40)}{(3 \times 2)} \right]} = \frac{36}{287}$$

Quantity III:

$$\text{Required probability} = \frac{30}{85} = \frac{6}{17}$$

So, Quantity I < Quantity II < Quantity III

100. (b) Total rotten candies in pouch B

$$= 84 \times \frac{25}{100} = 21$$

Total good candies in pouch B

$$= 84 - 21 = 63$$

And total rotten caramel flavor candies in

$$\text{pouch B} = \frac{21}{3} = 7$$

$$\text{Required probability} = \frac{{}^7C_1 \times {}^{63}C_1}{{}^{84}C_2}$$

$$= \frac{(7 \times 63)}{(84 \times \frac{83}{2})} = \frac{21}{166}$$

101. (e) In pouch D:

Orange candies = 15,

Pineapple candies = 15 - K,

Caramel candies = 10 + 2K

Coffee candies = 10 + K

Total = 15 + 15 - K + 10 + 2K + 10 + K

$$= (50 + 2K)$$

$$\frac{15}{(50 + 2K)} = \frac{1}{4}$$

$$\Rightarrow 60 = 50 + 2K \Rightarrow 2K = 10 \Rightarrow K = 5$$

102. (b) Ratio of time taken of (P + Q + R) and (Q + R) = 5 : 8

Ratio of efficiency of (P + Q + R) to that of (Q + R) = 8 : 5

Let the efficiency of (P + Q + R) and (Q + R) be 8 units/day and 5 units/day respectively.

So, efficiency of P = 8 - 5 = 3 units/day

$$\text{Share of R out of total share} = \frac{8000}{16000} = \frac{1}{2}$$

That means efficiency of R = (P + Q)

$$\text{Efficiency of R} = P + Q = \frac{8}{2} = 4 \text{ units/day}$$

So, efficiency of Q = 4 - 3 = 1 unit/day

Time taken by P and Q together to complete the job = Z days

Time taken by P, Q, and R together to complete the job = (Z - K) days

$$\text{Now, } Z = 2(Z - K)$$

$$\text{Value of } Z = 2K$$

So, time taken by P and Q together to complete the job Z = 2K days = time taken by R alone to complete the job = Y days

$$\Rightarrow \frac{Y}{2} = K$$

So, statement I is true.

II. If Z is half efficient that as that of S.

Value of K = 15

$$\text{So, } Z = Y = 2 \times 15 = 30$$

S is twice efficient as that of R, so time taken by S to complete the job

$$= \frac{30}{2} = 15 \text{ days}$$

This statement is false.

III. Z = Y (This statement is true.)

So, only I and III are true,

103. (b) Speed of car P = 4Z km/h

Distance between point A and point B

$$= Y \text{ km}$$

Let speed of car Q = a km/h

$$\text{Now, } (4Z + a) \times 10 = Y$$

$$\text{Value of } a = \frac{Y}{10} - 4Z$$

Also,

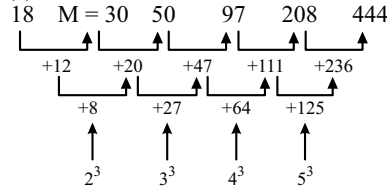
$$\left(\frac{3Y}{5} - 360 \right) = 6 \times \left(\frac{Y}{10} - 4Z - 2Z \right)$$

$$\Rightarrow \frac{3Y}{5} - 360 = \frac{3Y}{5} - 36Z$$

$$\Rightarrow Z = \frac{360}{36} = 10$$

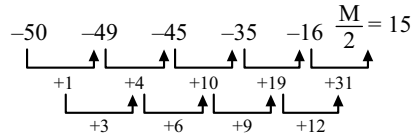
$$\text{So, speed of car P} = 4 \times 10 = 40 \text{ km/h}$$

104. (c) Pattern of series I:



So, value of M satisfies properly in I.

Pattern of series II:



So, value of M satisfies properly in II.

Pattern of series III -

$$8 \times 0.75 - 1 = 5$$

$$5 \times 1.5 - 1 = 6.5$$

$$6.5 \times 3 - 1 = 18.5 \neq M$$

$$18.5 \times 6 - 1 = 110$$

$$110 \times 12 - 1 = 1319$$

So, value of M not satisfies properly in III.

105. (d) Series I follow the pattern as:

$$12 \times 2 = 24$$

$$24 \div 3 = 8$$

$$8 \times 4 = 32$$

$$32 \div 5 = 6.4$$

$$6.4 \times 6 = 38.4$$

$$\text{So, } x = 32$$

Series II follow the pattern as:

$$11 + 9 = 20$$

$$20 + 13 = 33$$

$$33 + 17 = 50$$

$$50 + 21 = 71$$

$$71 + 25 = 96$$

$$\text{So, } y = 33$$

Series III follow the pattern as:

$$3 + 1! = 4$$

$$4 + 2! = 6$$

$$6 + 3! = 12$$

$$12 + 4! = 36$$

$$36 + 5! = 156$$

$$\text{So, } z = 36$$

Therefore, $x < y < z$

106. (b) I. Pattern of series:

$$P = 380 - (12^2 - 4) = 240$$

$$380 + (10^2 - 4) = 476$$

$$476 + (8^2 - 4) = 536$$

$$536 + (6^2 - 4) = 568$$

$$568 + (4^2 - 4) = 580$$

$$580 + (2^2 - 4) = 580$$

II. Pattern of series:

$$Q = 160 \times \frac{3}{2} = 240$$

$$240 \times \frac{4}{2} = 480$$

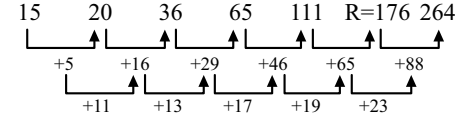
$$480 \times \frac{5}{2} = 1200$$

$$1200 \times \frac{6}{2} = 3600$$

$$3600 \times \frac{7}{2} = 12600$$

$$12600 \times \frac{8}{2} = 50400$$

III. Pattern of series:



So, P = Q > R will be true.

107. (b) A B 102 104 107 113 128 173

$$\begin{array}{ccccccc} & & +4 & +2 & +2 & +3 & +6 & +15 & +45 \\ & & \times 0.5 & \times 1 & \times 1.5 & \times 2 & \times 2.5 & \times 3 \end{array}$$

$$B = 102 - 2 = 100, A = 100 - 4 = 96$$

$$\text{Now, } M = 96 \times \frac{100 + 150}{100} = 240$$

Difference between M and B

$$= 240 - 100 = 140 \approx 144$$

$$\text{Required number} = \sqrt{144} = 12$$

108. (a) Let the speed of train A = x km/h

So, the speed of train B = (x + 25.2) km/h

Let the length of train B = 'd' m

So, the length of train A = (d + 45)m

According to question,

$$\frac{(d + 45 + 75) \times 18}{x \times 5}$$

$$\Rightarrow 5x = \frac{(d + 120) \times 18}{25}$$

... (i)

And,

$$\frac{(d+75) \times 18}{(x+25.2) \times 5} = 15$$

$$\Rightarrow (d+75) \times 18 = (5x+126) \times 15 \quad \dots(ii)$$

From equations (i) and (ii):

$$(d+75) \times 18 = \left[\frac{(d+120) \times 18}{25} + 126 \right] \times 15$$

$$\Rightarrow d+75 = \left[\frac{d+120+175}{25} \right] \times 15$$

$$\Rightarrow 5d+375 = 3d+885 \Rightarrow d = 255$$

From equation (i): $x = 54$

Speed of train A = 54 km/h = 15 m/s

Speed of train B = (54 + 25.2)

= 79.2 km/h = 22 m/s

Length of train A = (255 + 45) = 300 m

Length of train B = 255 m

Quantity I: Time taken by trains A and B to cross each other while running in opposite directions:

$$\frac{300+255}{15+22} = 15 \text{ seconds}$$

Quantity II: Original speed of train B = 22 m/s

So, new speed of train B = 22 + 3

= 25 m/s

$$\text{Required time} = \frac{255+95}{25} = 14 \text{ seconds}$$

Hence, Quantity I > Quantity II

109. (b) The speed of train b = 14 m/s

So, the speed of train A = $2 \times 15 - 14$
= 16 m/s

And, the speed of train C = $16 + 4 = 20$ m/s

Let the lengths of trains A, B and C are 'a'm, 'b'm and 'c'm respectively.

$$\text{Now, } \frac{a+b}{16+14} = 13 \Rightarrow a+b = 390 \quad \dots(i)$$

$$\text{Also, } \frac{a+c}{20-16} = 100 \Rightarrow a+c = 400 \quad \dots(ii)$$

Since, train A can cross a 140 m long platform in 20 seconds.

$$\text{So, } \frac{a+140}{16} = 20 \Rightarrow a = 180$$

From equation (i): $b = 210$

From equation (ii): $c = 220$

Quantity I: Since, original speed of train C = 20 m/s

So, new speed of train C would be

= 95% of 20 = 19 m/s

And, the time taken by train C to cross train B while running in the same direction

$$\text{would be} = \frac{210+220}{19-14} = 86 \text{ seconds}$$

Quantity II:

1st term of the series = 180

2nd term of the series = $180 - 2^2 = 176$

3rd term of the series = $176 - 3^2 = 167$

4th term of the series = $167 - 5^2 = 142$

5th term of the series = $142 - 7^2 = 93$

Hence, Quantity I < Quantity II

$$110. (b) \quad 3x - 4y = 2 \quad \dots (a)$$

$$x - 2y = 8 \quad \dots (b)$$

By equation (a) - equation (b) $\times 2$:

$$3x - 4y - 2x + 4y = 2 - 16$$

$$x \rightarrow -14$$

From equation (b): $y = -11$

Quantity I: Value of 'x' = -14

Quantity II: Value of 'y' = -11

Quantity III: Since MP of the article = ₹1000

So, CP of the article = $1000 - 250$

= ₹750

And SP of the article = $1000 - 200$

= ₹800

Now,

$$P = \frac{250}{750} \times 100 = 33\frac{1}{3}\%$$

And,

$$Q = \frac{200}{1000} \times 100 = 20\%$$

$$(Q - P) = 20 - 33\frac{1}{3} = -13\frac{1}{3}$$

Hence, Quantity II > Quantity III > Quantity I

111. (a) Logic in series A:

$$2^2 - 1 = 3$$

$$3^2 - 1 = 8$$

$$5^2 - 1 = 24$$

$$7^2 - 1 = 48(45)$$

$$11^2 - 1 = 120$$

$$13^2 - 1 = 168$$

So, wrong term of the series = 45

And the next term of the series

$$= 17^2 - 1 = 288$$

Logic in series B:

$$14 \times 2 = 28$$

$$14 \times 3 = 42$$

$$14 \times 5 = 70(56)$$

$$14 \times 7 = 98$$

$$14 \times 11 = 154$$

$$14 \times 13 = 182$$

So, wrong term of the series = 56

And the next term of the series

$$= 14 \times 17 = 238$$

Quantity I:

Average of wrong terms of both the series

$$= \frac{45+56}{2} = \frac{101}{2} = 50.5$$

Quantity II:

Difference between next terms of both the series

$$= 288 - 238 = 50$$

Hence, Quantity I > Quantity II

Solutions (112-115):

The number of candidates who got interview call for 6 posts in company P = $60 \times 6 = 360$

The number of candidates, who got interview call for 12 posts in company Q = $6x \times 12 = 72x$

The number of candidates, who got interview call for 2 posts in company R = $4x \times 2 = 8x$

According to the data provided in the question, we get

$$\Rightarrow 72x - 8x = 320 \Rightarrow 64x = 320 \Rightarrow x = 5$$

The number of candidates, who got interview call for 12 posts in company Q = $72 \times 5 = 360$

The number of candidates, who got interview call for 2 post in company R = $8 \times 5 = 40$

Let the total number of candidates applied in company Q = $4a$

$\therefore$ The total number of candidates applied in company R = 25% of $4a = a$

The total number of candidates applied in company P = $\frac{5a}{3}$

112. (d) It cannot be answered.

113. (c) Required percentage

$$= \frac{a}{\frac{5a}{3}} \times 100 = 60\%$$

114. (a) The total number of candidates, who got interview call in company P = 360

The number of female candidates, who got interview call in company P

$$= 360 \times 0.40 = 144$$

The number of females, who got job in

$$\text{company P} = \frac{1}{3} \times 6 = 2$$

The number of females, who did not get the job in company P = $144 - 2 = 142$

115. (d) According to question,

$$\frac{5a}{3} - a = 160 \Rightarrow \frac{2a}{3} = 160 \Rightarrow a = 240$$

The total number of candidates applied in company Q = $4 \times 240 = 960$

The number of candidates who did not get interview call from company Q

$$= 960 - 360 = 600$$

116. (b) The maximum limit for UPI123 Pay and UPI Lite Wallet transactions are ₹10,000 and ₹5,000, respectively.

UPI123 Pay is designed for feature phones, allowing users to make UPI payments without internet access. It works through IVR, missed calls, or app-based solutions on basic phones.

UPI Lite is an on-device wallet introduced to enable fast, low-value transactions without requiring a bank's real-time system every time.

117. (b) The implementation period for the National Green Hydrogen Mission (NHM) is 2023-2030.

The National Green Hydrogen Mission (NGHM) was launched by the Government of India on January 4, 2023.

Phase I focuses on creating demand, enabling supply through domestic electrolyser manufacturing, and initiating pilot projects in hard-to-abate sectors.

Phase II aims to make clean hydrogen cost-competitive.

- 118. (c)** Integral Coach Factory (ICF) in Chennai manufactures the Vande Bharat train in India.
ICF is a production unit under Indian Railways.
It developed the first Vande Bharat indigenously under the "Make in India" initiative.
The train features modern amenities, high-speed capability, and energy-efficient design.
- 119. (c)** The interest rate for the Sukanya Samriddhi Scheme for the quarter from October 1, 2024, to December 31, 2024 is 8.2%.
Sukanya Samriddhi Yojana is a government-backed savings scheme meant for the parents of girl children.
It is part of the Beti Bachao, Beti Padhao initiative and offers high interest rates with tax benefits, making it a great option for long-term financial planning.
Minimum deposit amount is ₹250 per year.
- 120. (d)** On 16th October 2024, HDFC Bank opened its first branch in Singapore.
This move aligns with the bank's goal to serve the global Indian diaspora and expand its international banking services.
This move supports HDFC Bank's vision to scale beyond Indian borders and offer world-class banking services internationally.
The branch opening followed regulatory approvals from Monetary Authority of Singapore.
- 121. (b)** The Return on Assets for Scheduled Commercial Banks as of March 2024 is 1.3%.
RoA is a key profitability metric that shows how efficiently a bank uses its assets to generate profit.
Scheduled Commercial Banks are banks that are listed under the Second Schedule of the Reserve Bank of India Act, 1934.
- 122. (a)** State Bank of India has recorded the highest profits is the FY 2024.
SBI's net profit for Q4 FY24 was ₹20,698 crore, showing a 24% year-on-year increase.
In comparison, other public sector banks reported lower net profits for the same period.
- 123. (c)** India's rank in the 2024 Global Nature Conservation Index (NCI) is 176th.
This ranking places India among the lowest in terms of nature conservation efforts, showing challenges in biodiversity preservation and environmental protection.
The Global Nature Conservation Index evaluates the effectiveness of nature conservation efforts across 180 countries, focusing on several key factors.
- In the 2024 rankings, Luxembourg secured the top position with a score of 70.8.
- 124. (c)** Shivraj Singh Chauhan is the current Union Minister of Agriculture and Rural Development.
Prior to his current role, Chouhan served as the Chief Minister of Madhya Pradesh from 2005 to 2018 and again from 2020 to 2023.
Shivraj Singh Chouhan is an Indian politician from Madhya Pradesh, affiliated with the Bharatiya Janata Party.
- 125. (d)** In November 2024, the Ministry of Railways planned to equip over 66,000 route kilometers (rkm) of its network with the indigenous Automatic Train Protection system named Kavach by 2030.
The Kavach system is an indigenous Automatic Train Protection technology developed by Indian Railways to enhance train safety by preventing collisions through continuous monitoring and automatic braking if necessary.
As of April 2022, Kavach was implemented on 1,445 route kilometers and 144 locomotives in the South Central Railway zone.
Approximately ₹50 lakh per route kilometer and ₹80 lakh per locomotive have been earmarked for the installation of Kavach.
- 126. (a)** The Paid-up Capital of the Reserve Bank of India is ₹5 crore.
This was set at the time of its establishment in 1935.
Initially, private shareholders held the majority of these shares, with the central government subscribing to a smaller portion.
- 127. (c)** Under Basel III, the minimum Common Equity Tier 1 (CET1) capital requirement for banks in India is 8%.
Regulators may impose a capital conservation buffer, typically set at 2.5% of RWAs, to ensure banks build up capital during periods of economic stability that can be used during periods of stress.
Banks are required to maintain a minimum CET1 ratio of 5.5% of risk-weighted asset.
In addition to the minimum CET1 requirement, banks should also maintain a CCB of 2.5% of RWAs, also in the form of CET1 capital.
- 128. (c)** The Hockey Women's Asian Champions Trophy 2024 was held at Rajgir, Bihar.
The tournament took place at the newly developed Rajgir Hockey Stadium.
India won the gold medal by defeating China 1-0 in the final match.
The Hockey Women's Asian Champions Trophy is a premier field hockey tournament organized by the Asian Hockey Federation.
- 129. (a)** The maximum limit of SLR in India is 40%.
SLR is the minimum percentage of a commercial bank's net demand and time liabilities that it should maintain in the form of liquid assets like cash, gold, or approved government securities.
SLR is an important monetary tool to ensure the solvency of banks and control money supply in the economy.
- 130. (d)** In India, the percentage set as the Priority Sector Lending goal for Small Finance Banks is 75%.
SFBs are required to allocate 75% of their lending to sectors categorized as priority by the Reserve Bank of India, such as agriculture, micro, small and medium enterprises, and others.
This is higher than the target for other commercial banks, which is typically 40% of ANBC.
Priority Sector Lending is an important initiative by the Reserve Bank of India to ensure that vulnerable and under-served sections of the economy get adequate credit support from formal banking institutions.
- 131. (a)** State Bank of India has recently unveiled a research study named 'Of the Women, By the Women, For the NATION'.
The study highlights the rise of 'Lakhpati Didis' - women SHG members whose income has reached or crossed ₹1 lakh annually.
Over 65% of SHG women have transitioned to become 'Lakhpati Didis', indicating rising financial independence and economic empowerment.
This study focuses on the transformative impact of SHGs on rural women and highlights SBI's role in supporting financial inclusion.
- 132. (c)** The Kavach system functioned as expected during the trial in 7 different emergency situations.
The system functioned as expected, demonstrating its effectiveness in enhancing railway safety.
Kavach uses an automatic train protection system that helps prevent collisions by automatically stopping the train if it detects an obstacle or if the train is about to breach signals.
Kavach is integrated with the signaling system of Indian Railways, ensuring that trains receive alerts about signal violations or other critical safety issues.
- 133. (b)** The RBI instructed the four NBFCs to cease and desist from sanctioning and disbursing loans to take as part of the imposed restrictions.
This action was taken due to concerns over their excessive interest rates and non-compliance with regulatory guidelines.

The restrictions were effective from October 21, 2024, but the companies were allowed to continue servicing existing customers and conducting collection and recovery processes.

134. (b) Llama 3.2 is the name of the new large language model (LLM) introduced by Meta at Meta Connect 2024.

This model represents a significant advancement in AI, capable of processing both text and images.

Llama 3.2 can interpret and generate both textual and visual content, facilitating applications such as real-time image analysis, visual search engines.

Recognizing the importance of mobile platforms, Meta has optimized lighter versions of Llama 3.2.

135. (b) The new position limit for trading members in index futures and options as announced by SEBI is ₹7,500 crore or 15% of total open interest, whichever is higher.

This change aims to improve market stability and ensure a more orderly trading environment.

Starting from April 1, 2025, SEBI will implement intraday monitoring of these position limits, though no penalties will be imposed for any breaches during this initial phase.

136. (a) India and Japan signed a Memorandum of Implementation for the co-development of the UNICORN mast.

This collaboration marks the first instance of co-development and co-production of defense equipment between the two countries.

The UNICORN mast integrates multiple communication systems into a single, streamlined design, enhancing the stealth characteristics of naval platforms by reducing their radar cross-section.

137. (c) The loan amount provided by the International Finance Corporation (IFC) to Axis Bank for blue finance and green projects in India was \$500 million.

This collaboration marks IFC's first blue investment in India and represents the largest green financing initiative by IFC in the country to date.

The loan is designed to fund investments in areas such as water and wastewater management, reduction of marine plastic pollution and restoration of marine ecosystems.

In addition to blue finance, the funding will support green projects, including the development of green buildings and other climate-friendly infrastructure.

138. (c) The total outlay approved by the Union Cabinet for the seven schemes aimed at improving farmers' lives is Rs. 14235.30 crore.

Allocated ₹2,817 crore, this mission leverages technology to improve farmers' lives, focusing on creating an Agri Stack and a Krishi Decision Support System.

139. (c) The prize money for the Booker Prize 2024 was £50,000.

The Booker Prize, established in 1969, is one of the most prestigious literary awards in the English-speaking world.

The prize is open to authors from the United Kingdom, the Commonwealth of Nations, Ireland, and Zimbabwe, though this has changed over time.

In 2019, the prize was opened to any English-language novel, regardless of the author's nationality, so long as it was published in the UK.

140. (c) The India Space Research Organization's Pushpak mission is associated with developing a reusable launch vehicle (RLV).

The Pushpak mission is part of ISRO's RLV Technology Demonstration Programme.

The main focus of the mission is to demonstrate key technologies such as hypersonic flight and autonomous landing, which are crucial for the development of a reusable launch vehicle.

141. (e) As per the guidelines of the Reserve Bank of India, the ongoing minimum percentage for the Capital to Risk-weighted Assets Ratio that Cooperative Banks need to hold is 9%.

This requirement ensures financial stability and protects depositors by making sure banks maintain sufficient capital relative to their risk-weighted assets.

Tier 2 to Tier 4 UCBs are required to maintain a minimum CRAR of 12%.

142. (d) 49.37% of the total order value on GeM during FY24 was contributed by Micro and Small Enterprise (MSE) sellers.

This is nearly half the total procurement volume, which not only highlights the growing participation of small businesses but also the government's push to empower MSEs through digital procurement.

This significant contribution underscores the key role MSEs play in public procurement through the GeM platform.

143. (e) Employees' Provident Fund Organization is not one of the promoters of the National Centre for Financial Education (NCFE).

The National Centre for Financial Education (NCFE) is promoted by the following financial sector regulators - Reserve Bank of India, Securities and Exchange Board of India, Insurance Regulatory and Development Authority of India and Pension Fund Regulatory and Development Authority.

EPFO is a statutory body under the Ministry of Labour and Employment, Government of India, and is not a financial sector regulator.

144. (c) Slice has received approval for its merger with North East Small Finance Bank.

The merger was completed on October 27, 2024, following approvals from the Reserve Bank of India, the National Company Law Tribunal, and the Competition Commission of India.

This strategic move allows Slice, a Bengaluru-based fintech startup, to transition into a full-fledged banking institution.

145. (d) Incorporation is not one of the five pillars of the Reserve Bank of India's (RBI) Payments Vision 2025.

The pillars aim to transform India into a global leader in digital payments by focusing on security, accessibility, technological advancement, strong institutional frameworks, and international reach.

The Reserve Bank of India's Payments Vision 2025 is a strategic framework released in June 2022, aiming to enhance the digital payment ecosystem in India.

146. (c) The expected launch date for India's Venus Orbiter Mission (VOM) unofficially referred to as "Shukrayaan" is March 2028.

The mission is expected to reach Venusian orbit on 19 July 2028, following a 112-day interplanetary journey.

The mission will be launched from Satish Dhawan Space Centre (SDSC), Sriharikota.

Launch Vehicle Mark-3 (LVM3) will be the launch vehicle for the mission.

147. (e) As per the Reserve Bank of India (RBI) regulations, the minimum Net Owned Fund (NOF) requirement for NBFC-Micro Finance Institutions (NBFC-MFIs) is ₹10 crore with effect from October 01, 2022.

This change was made to strengthen the NBFC sector and align it more closely with evolving risks and financial stability concerns.

148. (a) Kiran Rao is the director of the film "Laapataa Ladies".

This marks her return to directing after her debut with Dhobi Ghat in 2011.

The movie has received critical acclaim and won multiple awards, including Best Picture and Best Direction at the 2025 International Indian Film Academy Awards.

Born on November 7, 1973, in Bangalore, India, Kiran Rao is an Indian film director, screenwriter, and producer known for her significant contributions to Hindi cinema.

149. (a) Deakin University, Australia has become the first foreign university to receive approval of the International Financial Services Centres Authority.

Deakin's campus in GIFT City is set to offer the same degree programs as its Australian campuses, particularly in fields like science, technology, engineering, mathematics, and financial management.

This initiative aligns with India's broader strategy to attract top global universities to GIFT City.

150. (b) Maharashtra is the top performer in terms of the number of micro, small, and medium enterprises (MSMEs) in India as of March 2024.

With around 17% of the total MSMEs, Maharashtra is followed by Tamil Nadu.

The Ministry of MSMEs manages various schemes in order to provide credit and financial assistance to MSMEs.

151. (d) Statements 1, 2 and 3 are correct.

The Union Budget 2024-25 allocated ₹19,100 crore to the Ministry of New and Renewable Energy (MNRE), marking a significant increase from the previous year's revised estimate of ₹7,928.89 crore.

The 2024-25 Interim Budget provided for a budgetary allocation of ₹10,000 crore to solar power grid projects in FY2025 BE, which is a massive 110 per cent increase

from ₹4,757 crore allocated in FY2024 Revised Estimates, according to the Expenditure Budget document for 2024-25

The PM Surya Ghar Muft Bijli Yojana was launched with an investment of over ₹75,000 crore, aiming to provide rooftop solar installations to 1 crore households by 2027.

152. (c) 'Incoterms', the short form of International Commercial Terms, is a registered trademark of the International Chamber of Commerce (ICC).

These rules are widely used in international and domestic contracts for the sale of goods to clearly communicate the tasks, costs, and risks associated with the transportation and delivery of goods.

The International Chamber of Commerce is a global business organization that promotes international trade, responsible business conduct, and a global approach to regulation and policy.

It acts as a bridge between businesses and governments to facilitate global commerce.

153. (e) The "R" in LIDAR stand for ranging.

LIDAR stands for Light Detection and Ranging.

It's a remote sensing method that uses laser light to measure distances to a target.

The term Ranging describes the process of calculating the distance between the sensor and an object based on the time it takes for the light to bounce back.

154. (c) The Reserve Bank of India (RBI) has set the minimum paid-up capital requirement for establishing a Small Finance Bank (SFB) in India at ₹200 crore.

The capital must be in the form of paid-up equity capital.

Promoters must contribute at least 40% of the paid-up capital initially.

This move aims to strengthen the foundation of new SFBs and ensure their long-term viability.

155. (a) Network People Services Technologies Limited (NPST) has secured orders from two Kerala Gramin Bank and Karnataka Gramin Bank for its banking Superapp and digital transformation.

The SuperApp is designed to deliver modern, engaging banking and transactional experiences tailored to the needs of customers in rural and semi-urban areas.

NPST's platform integrates a wide array of services, including onboarding, everyday banking and payment transactions and credit products.

